

Online Information is Not Sex Education*

Laura Derksen (r) Catherine Michaud-Leclerc (r) Ardyn Nordstrom

May 2025

Abstract

Health misinformation can cause individuals to misjudge risks and lead to dangerous disease outbreaks. Misconceptions are widespread in the context of the HIV/AIDS epidemic. The internet has the potential to inform, but also to amplify misinformation. We show that access to online information, even restricted to a high-quality source, does not effectively counteract health misinformation. We study a randomized trial providing Wikipedia access in Malawian secondary schools. Despite intensive use, and nudges to search for correct information, nearly all students retain critical misconceptions related to pregnancy and HIV risk. Misconceptions persist for years after students gain access to the full internet. We find limited impacts on long-run sexual behavior and attitudes. High confidence in incorrect beliefs, and a preference for other online content, appear to explain the results. Access to online information does not substitute for effective sex education, and direct information provision is likely necessary.

*Derksen: Ragnar Frisch Centre. Contact: laura.derksen@frisch.uio.no. Michaud-Leclerc: Université Laval. Contact: catherine.michaud-leclerc@ecn.ulaval.ca. Nordstrom: Carleton University. Contact: ardynnordstrom@cunet.carleton.ca. We are grateful for insightful comments from Caitlin Brown, Pascale Dupas, Jonas Hjort, Jason Kerwin, and seminar and conference participants at McGill University, University of Oslo, AFE, CEA, and SMOQK. We thank Abdul Chilungo for project management assistance, and Alexander Russell Dore for research assistance. This study was approved by the University of Toronto Research Ethics Board, the Comités d'éthique de la recherche avec des êtres humains de l'Université Laval, the Carleton University Research Ethics Board, and the Malawi National Committee on Research Ethics in the Social Sciences and Humanities. The study was funded by SSHRC Insight Development and Partnership Engage grants. AEA Trial Registry ID AEARCTR-0003824. Author order randomized at https://www.aeaweb.org/journals/policies/random-author-order/search?RandomAuthorsSearch%5Bsearch%5D=Iw-uFEPo_f3L. All errors and omissions are our own.

1 Introduction

The spread of health misinformation poses a serious threat to individuals and society. Misinformation arises across contexts and with varied motivations, from deliberate campaigns to minimize the dangers of smoking, to flawed vaccine research and the propagation of fear-based HIV myths (Krishna and Thompson, 2021). Persistent false beliefs can deter people from taking the right precautions, increase health risks, and lead to dangerous disease outbreaks (Bursztyn et al., 2023). The internet mediates the spread of misinformation by providing users with access to a wide range of content, both true and false. Users can find detailed, accurate and up-to-date information on almost any topic, including health. On the other hand, the internet is rife with misinformation (Lazer et al., 2018; Carrieri et al., 2019; Acemoglu et al., 2024), which can spread with alarming speed during health emergencies (Do Nascimento et al., 2022).

In this paper, we study the impact of online information access on health misinformation in an important high-stakes context: the HIV/AIDS epidemic in sub-Saharan Africa (SSA). Misinformation is a particular threat to adolescent girls and young women in SSA, who face a dual burden of HIV infection and teenage pregnancy (Anderson, 2018; Karim and Baxter, 2019), both of which carry lifelong consequences and perpetuate cycles of poverty. Young people require accurate knowledge to make informed choices, and correcting common misconceptions about the risks of sex has been shown to impact both beliefs and behavior (Derksen et al., 2022; Kerwin, 2025), including for teenagers (Dupas, 2011a; Angrist et al., 2019). Yet, many adolescents in the region do not have access to comprehensive sex education (CSE). CSE is a subject of debate in the region, and many topics are viewed as controversial (Achen et al., 2024). Existing sex education classes lack detail and often focus on abstinence and resisting peer-pressure (Duflo et al., 2015; Sani et al., 2016). Rather than providing accurate information about risk, educators at times use fear-based messages that highlight the negative consequences of sexuality, which can breed misconceptions (Achen et al., 2024). Moreover, most adolescents in SSA miss out on sex education content delivered in secondary school, as 55 percent do not enroll (UNESCO, 2024). While most do not yet have internet access, internet use is rising quickly among young people across the region (International Telecommunication Union, 2025). If adolescents are introduced to (or restricted to) reliable online sources, they may be able to find the information they need.

We show that access to online information, even restricted to a high-quality information source, is not enough to combat critical and common misconceptions about sexual risk. We analyse data from a randomized controlled trial that provided Malawian secondary school students with access to Wikipedia, an online encyclopedia, during one school year. The intervention took place in government boarding schools where students had no other internet access. Treatment group students could access Wikipedia after

school and on weekends, on digital devices inside a classroom. We measure outcomes at the end of the school year, and in a long-run survey beginning five years after respondents had left school and gained access to the broader internet. At the end of the school year, we asked multiple-choice knowledge questions in two endline surveys, several weeks apart. For the long-run survey, we attempted to reach all treatment students and a randomized subset of control students, and achieved an 85 percent response rate. Both short and long-run surveys include the same knowledge measures, while the long-run survey adds measures of attitudes and behavior.

For our knowledge measures, we focus on four basic facts about pregnancy and HIV risk that are widely misunderstood, yet are critical in determining behavior and outcomes. First, we ask about the timing of the fertile window during the menstrual cycle. According to the [DHS \(2016\)](#), only 16 percent of Malawian women have accurate knowledge of the fertile period. Yet, this information is important for family planning in a setting where contraception is not always easily accessed. Second, we ask about the risk of HIV infection from unprotected sex with an infected partner. [Kerwin \(2025\)](#) finds that a majority of Malawian respondents believe the per-act infection risk to be 100 percent; the true risk is less than one percent ([Wawer et al., 2005](#)). [Kerwin \(2025\)](#) finds that providing accurate information on the risk of HIV transmission actually reduces risk taking for most respondents; those who believe HIV infection is inevitable (“fatalism”) tend to abandon caution. Third, we ask about *treatment as prevention*: the potential for HIV treatment to prevent transmission. A person who is treated for HIV is very unlikely to transmit HIV ([Cohen et al., 2011](#); [Rodger et al., 2016](#); [Bavinton et al., 2018](#); [Rodger et al., 2019](#)). Yet, in surveys conducted by [Derksen et al. \(2022\)](#) in rural Malawian communities, two-thirds of respondents were unaware that treatment could prevent transmission to any extent. [Derksen et al. \(2022\)](#) find that providing information on treatment-as-prevention increases HIV testing, which is critical for accessing HIV treatment. Finally, we ask about the risks associated with sexual partner choice. In Kenya, [Dupas \(2011a\)](#) surveyed teenage girls to find that only one-third of respondents understands that older men (over age 25) are more likely to be HIV positive than teenage boys. [Dupas \(2011a\)](#) shows that providing teenage girls with accurate risk information can affect sexual partner choice and reduce teenage pregnancy rates. [Angrist et al. \(2019\)](#) find similar results in the context of Botswana.

We find that Wikipedia access does not impact critical misconceptions about HIV and pregnancy risk, which are prevalent and persist over years. For each knowledge question at each survey point, most students in both the treatment and control groups answer incorrectly. Despite repeating the same questions over multiple surveys, and providing printed copies of the questions, students do not appear to be nudged into searching for answers. HIV-related misconceptions are particularly common. At the end of the school year, fewer than 5 percent of students answer any particular HIV question correctly. This improves only slightly after five to seven years (at most 15 percent), despite the fact

that both treatment and control students now have access to the full internet. Most respondents appear to believe fear-based messages about HIV: that unprotected sex leads to certain HIV transmission, even if a person is on HIV treatment, and that every potential sexual partner is equally high-risk. The impact of the intervention is estimated to be close to zero and is statistically insignificant for all health knowledge measures.¹ We finally observe a meaningful shift in beliefs after providing correct information directly to participants, during the long-run survey, sourced either from the United Nations (UN) or from social media. The UN information source was particularly effective, prompting the majority of respondents to report correct beliefs.

Beyond knowledge measures, we find limited impacts on long-run sexual health behavior, attitudes towards gender and sexuality, and internet use. We do not find any significant effect on unprotected sex in the past month or HIV testing. We find no change in attitudes towards lesbian, gay, bisexual and transgender (LGBT) people or gender equality, with the exception of a positive and significant impact on attitudes towards transgender people among female respondents. This estimate is not significant when adjusting for multiple hypothesis testing. In the long run, respondents in both the treatment and control groups are avid users of the internet, social media, and Wikipedia, with limited differences between the groups.

The stubborn persistence of health misinformation despite Wikipedia access, nudges to search, and several years of internet exposure is surprising for several reasons. First, students used Wikipedia intensively. During the school year the average treatment student spent a total of 22 hours on Wikipedia. One-fifth of that time was spent on pages linked to the school syllabus. Second, students clearly gained knowledge from Wikipedia, including on topics that were important for their academic success. [Derksen et al. \(2022\)](#) show an improvement in both English and Biology exam scores among low achievers, and use a range of tests to show that students can easily find information they need on Wikipedia.² Lastly, if students wanted to find sexual health information on Wikipedia it would not be difficult for them to do so. Correct answers to all of our knowledge questions appear on the relevant Wikipedia pages, and in general, students found it easy to search for, understand and trust the information available on Wikipedia ([Derksen et al., 2022](#)). Students were able to search for sensitive information anonymously, and spent 7 percent of browsing time on pages related to sex and sexuality. Indeed, the page for “Sexual Intercourse” was the most popular page overall.

Our findings appear to be explained by two key mechanisms. First, misconceptions are strongly-held: most students who answer the knowledge questions incorrectly are certain that they are correct, and therefore do not bother to search for information on

¹The study was designed with statistical power to detect a 6 percentage point increase in knowledge, and upper bounds based on 95 percent confidence intervals suggest that the intervention did not lead to a meaningful increase in knowledge among treated students.

²[Derksen and Souza \(2024\)](#) study the impact of the same intervention on social networks.

HIV risk. Second, Wikipedia offers an abundance of information, and sexual health topics do not capture students' attention to the same extent as topics related to sexuality, or other general interest topics. Treated students browsed primarily general interest pages (Derksen et al., 2022). Topic modelling indicates that the sex-related pages browsed by students were more often about sexuality than pregnancy or HIV; during the school year, the average student spent only 6 minutes on pages related to HIV. Most of the pages preferred by students were not of high quality. Indeed, control group students, who are not distracted by Wikipedia, are slightly more likely to learn about the fertility window (from their teachers and books) than treatment group students. It is possible that some students did search, but the information they found was difficult to understand or contradicted by other sources they trust. However, students report Wikipedia to be accessible and highly trustworthy, and use it to find factual information when prompted and incentivized (Derksen et al., 2022). Moreover, we would not expect students who receive conflicting or confusing information to report near-uniform beliefs with certainty.

While Wikipedia is a useful resource for students along other dimensions, it does not act as a substitute for effective sex education in the presence of misinformation. The broader internet is likely an even worse alternative, due to false and distracting content. Wikipedia does encourage students to read, which has been shown to impact academic performance (Derksen et al., 2022). It may also act as a useful complement to direct education campaigns by allowing users to verify new information. And, while students did not spend much time on pregnancy or HIV topics, they spent more time on sexuality topics, including on topics that are taboo in the Malawian context. This may carry benefits that are difficult to measure, for all students and in particular for LGBT students who may not be able to find relevant information locally.

This paper relates to a growing body of research that seeks to understand the causes and consequences of health misinformation, and interventions that counteract misinformation online. Several studies have documented negative consequences associated with health misinformation. Bursztyn et al. (2023) shows that health misinformation spread via ideological television news programming can lead to negative health outcomes, including death. Carrieri et al. (2019) show that misinformation about vaccinations online can decrease vaccination rates. Other work by List et al. (2024) and Pennycook et al. (2021) show that providing *targeted* interventions that encourage participants to change how they assess information can make them less susceptible to misinformation they encounter online, though these interventions are not specific to health information. We add to this research by showing that misinformation can persist over years, and that contrary to expectations, access to an information source is not enough to counteract health misinformation that is widely believed.

Our findings also present an important caveat to studies showing that internet access improves health knowledge. Internet access affects a range of outcomes in low-income

countries (Hjort and Tian, 2021). The internet reduces information frictions and increases knowledge in many domains (Agarwal et al., 2024; Porcher et al., 2024), including on the topics of HIV (Abbasi and Pongou, 2023) and contraception (Toffolutti et al., 2020) in SSA. Children and adolescents may however be prone to distraction when using the internet; Malamud et al. (2019) find no impact of home internet access on academic outcomes for Peruvian children. Yet, in previous work we found that Wikipedia access did increase both the ability to find accurate information and important academic knowledge, particularly for low achievers (Derksen et al., 2022). Taken together, our findings show that while students can and do use Wikipedia to find information they know they need, they do not use Wikipedia or the wider internet to check the veracity of strongly and widely held misconceptions.

Finally, this paper contributes to a broad literature on information interventions to improve health outcomes, by showing that simply providing access to an accurate information source may not suffice. In contrast, interventions that provide specific information can effectively correct misconceptions and impact behavior (Dupas, 2011b), including in the context of HIV (Dupas, 2011a; Angrist et al., 2019; Derksen et al., 2022; Yang et al., 2023; Ciancio et al., 2024; Kerwin, 2025). An “edutainment” approach, which embeds health information into an entertainment format, has also shown promise for improving both HIV knowledge and outcomes for young people (Banerjee et al., 2019). Education campaigns that promote preventative behavior directly, including through typical school-based sex education, have shown limited impacts in the context of HIV (McCoy et al., 2010; Duflo et al., 2015; Mwale and Muula, 2017) and in other health contexts (De Buck et al., 2017). More comprehensive sex education, even delivered through a digital platform, can effectively increase both health knowledge and demand for contraception (Chong et al., 2020).

2 Context and Experimental Design

2.1 Access to Information about Sexual Health in Malawi

Malawi is a low-income country in southern Africa, whose young people face many challenges related to sexual and reproductive health. Nine percent of the population is HIV positive, and one third of new infections occur among young people aged 15 to 24 (DHS, 2016). As of 2016, 64 percent of Malawian women had had sexual intercourse before the age of 18, and more than half became pregnant for the first time as teenagers. Only 15 percent of teenagers aged 15 to 19 use any form of contraception (DHS, 2016).

Sex education is part of the Malawian school curriculum in both the primary and secondary school, and is offered through the Life Skills class. The class was introduced in the 1990s as a response to the HIV/AIDS epidemic (Malata et al., 2023). The curriculum in-

cludes age-appropriate information on health promotion and diseases (including malaria, sexually transmitted infections, and HIV/AIDS), relationships and gender equality, culture and religion, self-esteem and personal development (Malawi Institute of Education, 2013). Importantly, the secondary school curriculum does not include specific information about risk reduction methods such as condoms or other forms of contraception. The curriculum focuses on “resisting pressure to engage in risky behaviors” (Malawi Institute of Education, 2013), suggesting a focus on abstinence and the extensive margin, rather than focusing on the intensive margin. The curriculum also exaggerates HIV infection risk. For example, the junior secondary school textbook contains a cautionary tale about a young woman who contracts HIV the very first time she has sex (Kadyoma et al. 2012, page 61).

School-based sex education excludes many of the most vulnerable young Malawians (Malata et al., 2023). While education attainment has progressed significantly in the past years, as of 2020 only 15.5 percent of Malawians had completed secondary school (UNESCO, 2024). According to the Malawi DHS (2016), the level of knowledge among young Malawians has remained essentially the same between 2010 and 2016, and most young Malawians have inadequate information on HIV prevention strategies.

Internet access is steadily increasing in Malawi and across SSA, particularly among young people, providing them with greater opportunities to access vital information. The share of the Malawian population that had used the internet in the past three months has risen sharply over the past decade, from 3 percent in 2011 to 24 percent in 2021 (see Appendix Figure A1). 3G and 4G networks are widely available, even in rural areas and during electricity outages (Batzilis et al., 2010). While internet infrastructure is present in many parts of the country, Malawian schools do not typically provide internet access to students. Mobile phones are usually prohibited, and computer labs, where available, are typically offline.

2.2 The Intervention

In 2017-18, we implemented an intervention which provided Malawian secondary school students with access to Wikipedia (Derksen et al., 2022). Wikipedia is a widely used reference site providing accurate information on almost every topic. It is an open source site, where content is created, verified, and updated at high frequency through open collaboration. Scientific content on Wikipedia is generally as accurate as, and more up-to-date than, content contained in offline encyclopedias (Giles, 2005). Information on Wikipedia is easy to find using a search function, and articles exist in both English and simple English (as well as in other languages). Wikipedia contains the information required to correct all of the misconceptions identified above as well as many others. The page devoted to the menstrual cycle contains text and illustrations to explain the fertile

window in detail, including its timing and duration. The HIV/AIDS page includes a table with average per-act HIV transmission rates, and states that “an HIV-positive person who has an undetectable viral load as a result of long-term treatment has effectively no risk of transmitting HIV sexually”.³

The study took place in four government boarding schools. Beyond the Life Skills class, the schools have limited informational resources on sexual and reproductive health, but did have small libraries primarily stocked with assigned text books. Students reside at the school during term and have no access to the internet outside of our intervention. The school grounds have consistent 3G or 4G network coverage. However, students are not allowed to have phones at school.

In each school, we set up a “digital library” in a school classroom, which would be open after school hours and during the weekends throughout the school year. Inside the digital library, students were able to browse Wikipedia using Android devices. The devices were equipped with a software to prevent students from accessing other websites. Students used the digital library under staff supervision, but were able to browse pages privately at their own desks. They were not authorised to talk or work in groups inside the digital library. Students could not leave the library with the devices, but they were allowed to take notes.

2.3 Experimental Design

At the start of the school year, students were randomly assigned to either a treatment arm, which gave them access to the digital library, or a control arm. The study included students in the last three years of secondary school (Forms 2, 3 and 4). Randomization was at the individual level and stratified by school, form, whether the student had ever used the internet, and whether the student’s baseline exam score was above the median.⁴ 301 students were assigned to the treatment arm and 1,207 students were assigned to the control arm. A subset of the control arm was randomly selected to complete additional surveys in both the short and long run. We refer to these 298 students, as well as the 301 treated students as the *survey sample* (599 students in total). Baseline characteristics for the survey sample are balanced (see Panel A of Appendix Table A1). As designed, the study had statistical power to detect a minimum increase in knowledge rates from 5 percent in the control group to 11 percent in the treatment group, and a 0.22 standard deviation change in HIV risk behavior.

Treated students were invited to an induction session to learn about the digital library, the use of Wikipedia-enabled devices, and privacy protections. Students drew a username

³https://en.wikipedia.org/wiki/Menstrual_cycle and <https://en.wikipedia.org/wiki/HIV/AIDS>, accessed on 16 October, 2024.

⁴The baseline exam score was calculated as the average of the student’s English and Biology scores (core subjects for which we have nearly complete data).

from a hat, which they would use to log in and browse Wikipedia anonymously. The username does allow us to track browsing behavior for individual students, and to associate behavior with some coarsened student characteristics.

Treatment students could browse Wikipedia inside the digital library after school and on weekends. The digital librarians maintained a waitlist, and device use was limited to 30 minutes during high-demand periods. The digital librarian was responsible for ensuring that only treated students could access the digital library by checking the student’s identity. This staff member also recorded arrival and departure times for each student, managed the Android devices, and supervised the use of the library.

3 Sex-Related Browsing on Wikipedia

During the 8-month period between November 2017 and June 2018, we tracked the time students spent in the digital library. Anonymized, individual-level browsing data was collected directly from the Wikipedia-enabled devices throughout the intervention. After students logged in, the devices monitored the Wikipedia pages they look at and the time they spend on each page. Each student spent an average of 22 hours browsing Wikipedia (6,674 hours in total) in the digital libraries and looked at 338 distinct Wikipedia pages (101,808 pages in total). There were no restrictions around what kinds of Wikipedia pages students could access.

Students spent approximately 7 percent of their time on pages related to sex and sexuality (Derksen et al., 2022). Our analysis focuses on these pages, using the same classification as in Derksen et al. (2022) to identify pages related to sex and sexuality. Since each page passes through multiple paths in Wikipedia’s category tree, we categorize pages as sex-related if 10 percent or more of these paths go through Wikipedia’s subcategories for “human reproduction”, “human sexuality”, “sexual health”, and “sexuality and society”.

3.1 Topic Modeling

To examine the content of the pages students looked at, we capture the introduction text from each page. The page details are based on the content in Wikipedia as of January 2024, since historical data cannot be consistently collected.

Restricting our attention to the set of the 3,407 sex-related pages viewed by the students, we classify the content of each page introduction using topic modeling. Topic modeling refers to a broad class of machine learning methods that categorize text-based documents—in this case, Wikipedia introductions—into groups based on how often the terms in those documents occur together. Topic models have been used widely in other social sciences, such as political science (Grimmer and Stewart, 2013; Jelodar et al., 2019), as well as a handful of studies in economics (Gentzkow et al., 2019; Larsen and Thorsrud,

2019; Ash et al., 2025; Ferrario and Stantcheva, 2022). These models are based on the premise that documents are mixtures of topics, and topics are mixtures of words. To determine which words fit within a topic, we compare how often words are used within a particular introduction compared to how often they are used across all introductions (within the set of pages related to sex).

We use a Latent Dirichlet Allocation (LDA) model to identify the main topic in each of the page introductions. LDA is an unsupervised method that identifies words that are more likely to be associated with one another than with other words in the “corpus” of documents (in this case, the corpus refers to the set of all sex-related Wikipedia introductions viewed by students). This model assigns each word in the corpus with a probability that it is associated with each topic (where the total number of topics, K , is fixed). This probability can then be used to predict the primary topic each page belongs to (Jelodar et al., 2019). The assignment to topics is done using the “TextmineR” package in R (Jones et al., 2021), which follows the process:

1. Assign each instance of any word (a *token*), across all introductions, randomly and independently, to exactly one out of K topics.
2. Count how often a given word (e.g. “woman”) falls into this topic, across all introductions. This will produce a probability distribution over *words* for each topic and count how often each topic appears within each introduction.
3. Reassign each token, across all introductions, using Gibbs sampling by iteratively doing the following steps:
 - Remove a token from its current topic, to compute the new probability distributions.
 - The removed token is then reassigned to a topic where (i) the word is prevalent in the topic and (ii) the topic is prevalent in the introduction the token belongs to. All counts and probability distributions are then recalculated.
 - Tokens are continuously reassigned until a stable distribution of topics over words is identified.

The result of this iterative process is to find a set of stable topics, where each topic is characterized by a distribution over words. The number of topics K has been selected based on the optimal level of “coherence,” which measures topic quality and reflects how often words in a topic are used together (Jones et al., 2021).⁵ We restricted the number

⁵The specific measure used here is calculated as the probabilistic coherence, which estimates the difference between (i) the likelihood that a particular word is included in introductions which also include the other words in the topic and (ii) how frequently that word appears across all introductions (Jones, 2019). A higher topic coherence value implies that words associated with that topic are more likely to be used within that topic than in the overall corpus of page introductions.

of possible topics to between 2 and 25 for interpretability and find that with 14 topics we achieve the highest level of coherence across topics. Since each topic is a distribution over words, several topics can contain the same word if these words appear frequently across multiple topics. In our case, the pages in the corpus have been restricted to those related to sex and sexuality, so the word “sex”, for example, appears in multiple topics.

The top terms associated with each of the topics are described in Table 1. To remove human judgment from the labelling process, the topic labels have been assigned using ChatGPT based on the Top 10 words from each topic.⁶

3.2 Browsing Time and Quality

To assess the quality of information that students in the treatment group were exposed to, we supplement the browsing data from the original experiment with information from each Wikipedia page that appears in the browsing data. Using the “WikipediR” package in R, which provides a wrapper of the Wikimedia Application Programming Interface (API) portal (Bar-Hen et al., 2016), we collect information about the content and quality of each page. The quality measures are based on Wikipedia’s page content assessment ratings and provide a measure of an article’s completeness, which are assigned primarily by Wikipedia editors. Pages of B-Class and above are deemed to be suitably referenced and reasonably well-written. This meets most of Wikipedia’s content suggestions, so we assign any page with a category of B-class or higher as being “good” quality (Wikimedia Foundation, 2023).⁷

Sex-related page visits and browsing time are concentrated primarily on lower-quality pages. Figure 1 shows the average time spent on each topic per user (by gender) and the average page quality for pages in different sex-related topics. 15 percent of the 3,407 sex-related pages browsed by students meet the B-class threshold for being of “good” quality, and students spent only 39 percent of their time on good-quality pages. As a point of comparison, Wikipedia pages related to the school syllabus are of higher quality than the sex-related pages browsed by the students. The horizontal line in Appendix Figure A2 shows that 37 percent of all syllabus-related pages meet the good-quality threshold. The subject with the lowest share of good pages is social sciences, where 26 percent of pages are of good quality. Within syllabus-related browsing, the average student spent 58 percent of their time on good-quality pages.

The darker portion of the bars in Figure 1 shows the proportion of time students spend on good-quality pages within each topic. Low-quality browsing is common within topics that are more clearly related to entertainment, such as “sexual activities and behaviors”, “pornography and adult magazines”, and “genital anatomy and function”. Students spent the most time on pages related to “sexual activities and behaviors”. The average

⁶See chat here: <https://chatgpt.com/share/4e6bdf59-9a0b-4da2-b2a0-8b9d26a42723>.

⁷For more information on the content assessments, see [Wikipedia’s Content Assessment ratings](#).

student spent 54 minutes on this topic, and 24 minutes on good-quality pages within this topic. Topics related to sexual health, such as “women’s reproductive health”, “pregnancy and childbirth”, and “HIV/AIDS and infection” are higher quality both in terms of the proportion of good-quality pages and time spent. For example, 32 percent of pages in the “HIV/AIDS and infection” topic are B-class or better. However, these topics received less attention from students. The average student spent only 8 minutes on “women’s reproductive health”, 7 minutes on “pregnancy and childbirth”, and 6 minutes on pages related to “HIV/AIDS and infection”.

Overall, male students spent more time than female students browsing sex-related pages. Both male and female students spent the most time on pages related to “sexual activities and behaviors”, with male students having spent more time on pages related to “pornography and adult magazines”, “genital anatomy and function”, and “BDSM and fetish culture”. Female students spent slightly more time on pages related to “HIV/AIDS and infection”.

3.3 Sexuality Diversity

Beyond information about sexual health, access to online information also provides students with other helpful information they cannot find elsewhere. This may be particularly valuable for students who identify with the LGBT community, as sexuality diversity is a sensitive and/or taboo topic in many cultural contexts. This is particularly true in Malawi, where same-sex relationships are illegal, LGBT visibility is low, and LGBT youth are subject to social isolation ([Angotti et al., 2019](#)). Figure 2 shows that students who were more socially isolated spent significantly more time looking at pages related to “gender and sexuality diversity” than students who were less socially isolated. Here, social isolation is defined as having a below-median number of friends (total number of social links) as recorded in the baseline survey.

4 Impact on Knowledge and Behavior

The preceding discussion described how students used Wikipedia to browse information about sex and sexuality-related topics. In this section, we describe the data we use to capture knowledge and behavior outcomes related to pregnancy and HIV, our empirical strategy to estimate the impact of Wikipedia access, and our main results in the short and long run.

4.1 Data

Our outcomes are captured in short-run (endline) and long-run survey data collected from study participants in both treatment and control groups. The survey sample includes 599

students in total, as described in Section 2.3. We conducted two closely-spaced in-person endline surveys at the end of the school year: one in May 2018 and another two to five weeks later.⁸ We also conducted a long-run follow-up survey between 2023 and 2025, in which we attempted to reach former students five years after the intervention took place. This long-run data collection involved recontacting participants who had since graduated from secondary school, using the contact details they provided in the original survey, and through social networks as needed. Most interviews took place over the phone, and some were conducted in person.

During the first endline survey in 2018, we asked students three multiple choice questions to capture knowledge related to the common sexual health misconceptions presented in Section 2.1. First, we asked about the length of a woman’s fertile window during each menstrual cycle. Second, we asked about the HIV transmission rate. Third, we asked about the effectiveness of antiretroviral treatment (ART) in preventing HIV transmission. The wording of the questions and multiple choice options are presented in Appendix Table A2. Young people also reported how certain they were about each answer. After the first endline survey, we gave all students a piece of paper with the questions and multiple choices printed out.

During the second endline survey, we repeated the same three questions. Young people were not incentivized to find the right answer between the two endline surveys, and we did not tell them that we would repeat the questions. During this second endline, we added a fourth question addressing another common misconception about the likelihood of HIV-infection for men in different age groups. We again asked students how certain they were about their answer. In this second endline survey, we also asked students where they had learned the information.

The long-run follow-up survey covered a large number of outcomes. We repeated the four knowledge questions described above, as well as the certainty measures. Since these knowledge questions may not capture all of the relevant information young people use to inform their behavior, we added several new questions about sexual health behaviors. The long-run survey also included questions about attitudes towards gender norms and rights, access to the internet, and internet usage patterns.

We observe attrition at each follow-up survey. At the first endline, 95.2 percent of study participants responded to the survey. 91.7 percent responded to the second endline. 543 participants responded to both endline surveys (90.7 percent). In the long-run follow-up survey, 85.3 percent of the young people responded to the questionnaire (out of 599 potential respondents). We do not observe differential attrition by treatment status (see Panel B of Appendix Table A1).

⁸About 5 percent of interviews took place less than two weeks after the first endline and less than 20 percent of interviews occurred six to nine weeks after.

4.2 Empirical Strategy

Our main regression specification for evaluating the short-term impact of Wikipedia access on health knowledge is as follows:

$$y_i = \alpha + \beta T_i + \sigma_s + \varepsilon_i \quad (1)$$

where y_i is the outcome of interest of individual i , α is a constant, T_i is a dummy variable indicating the treatment status, and σ_s is a stratum fixed effect. Our coefficient of interest is β , the estimated effect of Wikipedia access on the outcome of interest. ε_i is the error term. We present robust standard errors (Bruhn and McKenzie, 2009). To account for multiple hypothesis testing, we also include q-values following Anderson (2008), which controls for false discovery rates.

We also estimate heterogeneous treatment effects by gender in which we interact the treatment dummy variable with dummy variables that equal 1 if i is a male ($Male_i$) or female ($Female_i$) respondent:

$$y_i = \alpha + \beta^M T_i \times Male_i + \beta^F T_i \times Female_i + \delta Male_i + \sigma_s + \varepsilon_i \quad (2)$$

The analysis of the long-run data follows the same specification as equation 1, but includes a continuous time trend t_i capturing the month and year of the long-run follow-up survey, as follows:⁹

$$y_i = \alpha + \beta T_i + \sigma_s + t_i + \varepsilon_i \quad (3)$$

Our treatment effects are estimated following an analysis plan, with deviations clearly noted in the text. The analysis plan was prepared before long-run data collection was completed, but after short-run data was collected and analyzed for a separate paper (Derksen et al., 2022).¹⁰

4.3 Impact on Health Knowledge

Health Knowledge Outcomes

In the short run, we find that the vast majority of students in both the treatment and control groups remain misinformed about critical facts related to pregnancy and HIV risk. Panel A of Figure 3 shows the percentage of students who answer each sexual health question correctly, at the first endline survey, by treatment group. Overall, the great

⁹We pre-specified a specification that includes a time trend control as the follow-up data collection took approximately two years to complete. However, this control is potentially causally affected by treatment status. We include estimates of our main effects without the time trend in the appendix.

¹⁰We have also included descriptive facts and figures not specified in the analysis plan. Analysis plan available at <https://www.socialscisceregistry.org/trials/3824>.

majority of respondents in the treatment and control groups answer incorrectly. This is particularly true for the common misconceptions surrounding HIV: for each of these questions, fewer than 5 percent of students answer correctly. Students are more informed when it comes to the Fertility question; 31 percent of control students answer this question correctly.

Access to Wikipedia does not significantly improve short-run knowledge of these key facts. Table 2 presents the point estimates from equation 1. Estimated effects are null or negative for every knowledge question. Even taking 95 percent confidence intervals into account, it is unlikely that Wikipedia access leads to a meaningful number of students becoming informed. In the short-run, we can bound the effect size below 1 percentage point for knowledge of the true HIV transmission rate, and below 5 percentage points for knowledge about HIV treatment as prevention. We find no short-run impact of Wikipedia access on a knowledge index. The Knowledge Index is constructed using all three or four knowledge questions, and following [Anderson \(2008\)](#). We present the results in Columns 1 and 3 of Table 3, which shows no impact on knowledge at the first endline.

Low levels of knowledge persist even after students are given a nudge to search for the answer. At the first endline survey, students were given a printout of the knowledge questions. Panel B of Figure 3 and Table 2 present the results for these same questions two to five weeks later, as well as the percentage of students who know that older men are more likely to be HIV positive (i.e., the Sugar Daddies question, which was added to the second endline). Again, for the HIV questions fewer than 5 percent of students answer correctly.

In fact, female students in the control group are slightly more likely to learn the correct answer to the Fertility question between the first and second endline surveys than treatment students. Column 4 of Panel B in Table 2 shows that on average, relative to female students in the control group, female students with access to Wikipedia are significantly less likely to know the number of fertile days in an average menstrual cycle. Male students are also less likely to know the correct answer, but the difference is not statistically significant. While we also find a statistically significant negative effect in Column 7 (the Sugar Daddies question), only 2 percent of students (i.e., 6 students out of 549) answer the question correctly. We also estimate a negative and significant coefficient on the knowledge index at the second endline, (Column 3 of Table 3). In Figure 4 we show that none of these estimates, nor any other treatment effect estimates reported in the paper, are statistically significant after adjusting for multiple hypothesis testing ([Anderson, 2008](#)).

The limited sexual health knowledge we document is remarkably persistent over time, even after participants gain access to the full internet. Five to seven years after the endline, we asked the same four knowledge questions to the young people in the survey sample. Panel C of Figure 3 shows a similar pattern to Panel B (the second endline survey),

with the great majority of respondents answering incorrectly. We do see a slight rise in knowledge about HIV treatment as transmission, a subject of recent global public health campaigns¹¹, in both the treatment and control groups over time. Table 4 presents the long-run point estimates for each question in Columns 1 to 4, as well as for the Knowledge Index in Column 5. All treatment effect estimates are statistically insignificant.

In contrast, providing correct information to participants does appear to improve knowledge, particularly if the source is credible. Soon after eliciting beliefs in the long-run survey, we provided participants with the information they would need to answer two of the HIV questions correctly. To provide information on treatment as prevention, we shared this statement: *I want to talk about HIV and viral load. Viral load refers to the amount of HIV virus in an infected person's blood. Higher viral load usually means that the infection is worse, and a person with a high viral load may be very sick. ARVs can reduce a person's viral load, and if a person takes ARVs properly, the virus can even become undetectable. That means that blood tests will find extremely low levels of virus in the person's blood. According to the United Nations, people with HIV who achieve an undetectable viral load through ARVs – it is impossible for those people to transmit HIV.* For the HIV transmission rate question, we shared a statement posted “by an anonymous Twitter user”: *the truth is that it's not nearly that easy to get HIV – the medical literature estimates that the transmission rate is actually about 0.1% per sex act.* We elicit beliefs immediately after providing the information, and again after a 15 minute survey break. Figure 5 shows the percent of participants who answer correctly after being provided with correct information from the high-quality source (the UN, Panel A) versus from social media (Twitter, Panel B). In both cases, we see a significant jump in correct answers immediately after the information is provided, and a slight rise after the break. However, only the high-quality source persuades a majority of participants: 53 percent answer the question about ART correctly after information is provided, while the social media post increases the rate of correct responses from 2 percent to only 13 percent (15 percent after the break).¹²

Certainty and Misinformation

Respondents' answers to the knowledge questions suggest that beliefs about HIV are subject to specific, common misconceptions. Figures 6 to 9 show the exact multiple-choice answers chosen by the respondents. For the Fertility question (Figure 6), the most common choice is the correct answer at every time period. However, for each HIV-related question, the majority of respondents choose the same incorrect answer. At every survey

¹¹See, for example, <https://preventionaccess.com/>.

¹²We phrase the ART knowledge question slightly differently after the information provision to better match the UN statement: “Do you believe that it is impossible for a person with an undetectable viral load to spread HIV?”

point, approximately two-thirds of respondents believe the HIV transmission rate to be 100 percent, with most remaining respondents choosing 50 percent (Figure 7). In the short run, approximately four-fifths of students believe that HIV treatment has no effect on HIV transmission (Figure 8). And, in both the short and long run, the majority of respondents believe that older and younger men are equally likely to be HIV-positive (Figure 9).

Both treatment and control respondents report a high degree of certainty even over incorrect responses (see Appendix Table A3). Certainty over incorrect answers is high at the first endline (Panel A) and remains high at the second endline (Panel B) and the long-run follow-up survey (Panel C). HIV-related misconceptions are particularly strongly-held; for incorrect responses at the second endline survey, the respondent is *certain* or *very certain* 96 percent of the time. This is less true of the Fertility question, but most incorrect answers are still stated with certainty: 88 percent for girls and 97 percent for boys. Appendix Figures A3 to A5 present the source of information used to answer the knowledge questions, as reported by treatment and control students. For the Fertility question, both correct and incorrect answers come from a range of sources including teachers, the library, and Wikipedia. For the HIV questions this is also the case, but the most common source for both treatment and control students is a “guess”.

Wikipedia access does not appear to affect the likelihood that a student has strongly-held but misinformed beliefs. We construct a Misinformation Index using indicator variables that equal one when the response is incorrect and the respondent expresses certainty about their incorrect answer. As seen in Columns 2 and 4 of Table 3 and Column 6 of Table 4, the treatment effect estimates are negative in both the short term and long term.

4.4 Long-Run Impact on Sexual Behavior and Attitudes

Wikipedia access during secondary school may impact later sexual behavior and risk taking through various channels. In the previous section, we showed that Wikipedia access did not increase sexual health knowledge as measured by four key facts. Yet, the intervention may have affected knowledge or attitudes along other dimensions, with implications for behavior. In this section, we examine whether access to Wikipedia affected sexual behavior, attitudes, and internet use in the long term.

Sexual Health Behavior

We find no long-run effect of Wikipedia access across many behaviors known to be important for positive sexual health outcomes, including teenage marriage and pregnancy, unprotected sex, and HIV testing. Table 5 presents treatment effect estimates for (1) having sex before age 18, (2) marriage before age 20, (3) becoming a parent before age 20, (4) unprotected sex in the past month, and (5) HIV testing in the past year. We also

construct an index (6) that includes all sexual behaviors from outcomes in Columns 1 to 4. Most point estimates are very close to zero overall, and for both male and female respondents separately. For example, we find a less than 1 percentage point increase in the probability of unprotected sex in the past month, from a control group rate of 38 percent, and with confidence interval $(-0.08, 0.10)$. This indicates that the intervention is highly unlikely to meaningfully reduce risky sexual behavior in the long run. We also find no significant impact on overall HIV risk behavior. Column 7 of Table 5 shows the effect on the HIV Risk Index, which includes the number of sexual partners in the past month, the total number of partners ever, and an indicator for unprotected sex in the last month. Male respondents in general are more likely to engage in risky behavior, but we do not find a significant treatment effect for either gender; the average treatment effect estimate is -0.04 standard deviations. The only significant estimates in this table are a negative effect on the likelihood of sex before age 18, which contributes to a negative effect on the summary index measure in Column 6. These effects are entirely driven by the treatment effect on boys, who are slightly more likely overall to report having sex for the first time before 18 (Panel B of Table 5). These treatment effect estimates are not significant once we adjust for multiple hypothesis testing (Figure 4).¹³

Attitudes Towards Sexuality and Gender

In Section 3 we saw that treatment students spent 7 percent of their Wikipedia browsing time on topics related to sex and sexuality, including sexual activities and behaviors, gender and sexuality diversity, and sexuality and legal aspects. We might therefore expect Wikipedia access to impact long-run attitudes towards sexuality and gender.

Overall, we find no significant long-run impact on attitudes towards LGBT people, progressive gender norms, or sex before marriage. Table 6 reports the treatment effect estimates on several indices that capture (1) whether respondents can correctly define what it means for someone to be LGBT or transgender,¹⁴ (2) acceptance of same-sex relationships, (3) acceptance of transgender identity, (4) gender equality, and (5) acceptance of sex before marriage. All of the average treatment effect estimates are insignificant. However, female respondents in the treatment group are significantly more accepting of transgender people, and the point estimates on same-sex acceptance, gender equality, and

¹³The coefficients may be negative in part due to random imbalance: we observe that students in the treatment group were marginally less likely to have had sex before the intervention began (6, compared to 16 in the control group). Appendix Table A4 disaggregates findings by whether the school was co-educational or single-sex. Across school types, sex before age 18 occurs at a similar rate (perhaps outside of term time), and the treatment effect estimates are also similar.

¹⁴Whether the respondents knew the definition of the acronym LGBT was recorded by the enumerators themselves. For the definition of transgender, we asked enumerators to provide the definition given by the respondent. The researchers manually assigned a score based on their answers, based on approximate adherence to the Merriam-Webster dictionary definition: “of, relating to, or being a person whose gender identity differs from the sex the person was identified as having at birth”. We provide examples of respondents’ definitions in Appendix Table A5.

attitudes towards sex before marriage, while insignificant, are large and positive. Again, all point estimates are statistically insignificant when adjusted for multiple hypothesis testing (Figure 4).

Given that Malawian laws prohibit same-sex relationships, young people may be hesitant to express support for LGBT people. Indeed, more than three-quarters of respondents state that being in a same-sex relationship, or being transgender is never justifiable (Panel A of Appendix Figure A6).¹⁵ We observe substantially more variation in responses to questions about discrimination and criminalization of LGBT people, as well as abortion and gender roles (Panel B of Appendix Figure A6).

Long-Run Internet Use

Finally, we examine whether the Wikipedia intervention led to long-run differences in internet use patterns. Table 7 shows treatment effect estimates on access to smartphones, internet use, Wikipedia use, and use of social media and other sources for health information.¹⁶

Overall, we find that the intervention had few long-lasting impacts on the use of smartphones, Wikipedia or the internet more broadly. Almost all of the treatment effect estimates are insignificant overall and across genders. While the treatment group is significantly more likely to report having ever used Wikipedia, 86 percent of the control group also reports having done so, and in both groups the average student visited 5 Wikipedia pages in the past week (based on self-reported browsing history). Treated students are slightly more likely to use Wikipedia for health information (0.07 standard deviations), and less likely to use social media (0.05 standard deviations), but these differences are not statistically significant. Social media use in general is similar across treatment and control groups. We do find that women in the treatment group are significantly more likely than women in the control group to access free internet (an outcome that was not pre-specified).¹⁷

¹⁵In the analysis plan, we hypothesized, based in part on the browsing data (see Figure 2), that those who were more socially isolated during school may be more accepting of LGBT people later on. However, we do not find any significant correlation between having a below-median number of friends at baseline and LGBT acceptance (Appendix Table A6).

¹⁶Most of the outcomes in this table correspond with the outcomes specified in the study’s analysis plan. However, there are two key changes. First, the outcomes for “has digital wellbeing”, “free internet”, and “has browsing history” were not pre-specified, but have been included after data exploration. Second, the “Social Media and Health” and “Wikipedia and Health” outcomes in Panel D were pre-specified to include the ranked position of social media or Wikipedia as inputs into their respective indices. This has been modified to include instead an indicator for whether this source was included as one of their top three sources, as many higher rank positions were missing due to a data collection error. This change has no impact on the findings, but makes the index more informative as we have fewer missing values.

¹⁷Appendix Tables A7 to A10 present the long-run results excluding the time trend. All results are robust to excluding this control.

4.5 Mechanisms

Despite having access to an online information source with accurate, up-to-date health information, the students in our study did not gain critical knowledge required for family planning or to understand HIV infection risks. In this section, we shed light on two important mechanisms that may explain why Wikipedia access is not effective in improving sexual health knowledge or behavior. First, because sexual health misconceptions are believed with a high degree of certainty, students may not bother to search for new information online. Second, even when restricted to Wikipedia, students prefer to spend their time reading lower-quality, but perhaps more entertaining content, and are more interested in sexuality than in sexual health. The limited knowledge gains we document do not appear to be driven by limitations of Wikipedia as an information source or by limits of understanding. On the contrary, students find Wikipedia to be engaging, easy to use, and trustworthy.

Confidence in Common Misconceptions

The knowledge questions we posed are subject to widespread and strongly-held misconceptions, which may prevent people from seeking out true information. Across all three HIV questions, nearly every respondent consistently gets the answers wrong (Figure 3). Most of the incorrect answers are clustered around common misconceptions, including the misconception that HIV transmission is highly likely in a single act, and that every potential sexual partner is equally high-risk. These misconceptions are aligned with public health campaigns and school-based sex education programming which often emphasize HIV risk, as discussed in Section 2.1.

The vast majority of respondents in our study are confident that they already know the answers, even when incorrect. At the first endline survey, over 95 percent of incorrect respondents are certain they are correct (Appendix Table A3). After being prompted with the questions and receiving them in printed form, the students had several weeks to seek out the correct answers should they wish to. Yet, the pattern of incorrect responses and certainty is nearly identical at the second endline. Even at the long-run follow-up, 87 percent of incorrect respondents are certain they have the correct answer. This suggests that rather than not having considered these questions, people simply believe they already know the answers. Information about the fertile window, on the other hand, does not appear to be subject to a specific misconception. Respondents are generally more informed on the topic of fertility, and those who are misinformed at the first endline survey are slightly less certain. Some students in the control group improved their fertility knowledge between the two short-run endlines, suggesting that young people may be more inclined to seek out answers to questions when they are less certain.

Providing correct information directly does appear to shift misconceptions, partic-

ularly when arising from a credible source. During the long-run survey, most students appear to genuinely believe the UN statement on HIV treatment as prevention, as recorded in repeated survey measures and high certainty scores (see Panels E and F of Table A3). The rate of correct answers and certainty scores rise further after students are given a 15 minute survey break. Participants were asked why they did or did not change their answers (Appendix Figure A7); the terms “searched”, “internet”, and “googled” are relatively more important in responses from people who update their answers to be correct after the break.¹⁸ This suggests that the internet may act as a complement rather than a substitute for campaigns to combat misinformation directly.

Distraction Online

The fact that we do not see a positive impact on the fertility question, which is not as prone to strongly-held misconceptions as the HIV questions, suggests a second mechanism: students are drawn to Wikipedia pages about sexuality and other general interest topics, many of which are low quality, rather than sexual health information pages. Indeed, Wikipedia may distract students and take time away from other types of learning. Between the first and second endline survey, female students in the control group were more likely to find the correct answer to the Fertility question than female treatment students, apparently by asking their teachers (Table 2 and Appendix Figure A3).

During the intervention, the average treatment student spent a great deal of time browsing Wikipedia (22 hours), and 78 percent of that time was spent on general interest topics rather than school syllabus topics (Derksen et al., 2022). Indeed, Derksen et al. (2022) find that while the intervention impacted English exam scores by encouraging reading, most other schools subjects were not impacted. The average student spent 7 percent of browsing time on pages related to sex and sexuality, yet very little time was spent on pages directly related to pregnancy or HIV. Students spent more time on sexuality topics such as sexual activities and behaviors, pornography and adult magazines, and genital anatomy and function (Figure 1).

Students spent a disproportionate amount of sex-related browsing time on lower-quality pages (61 percent). This is not true of school-related browsing; for pages related to the syllabus, the average student spent 58 percent of time on good-quality pages. Moreover, the average student spent 2.5 hours on Biology pages, a common favorite school subject, and a subject for which Derksen et al. (2022) do find improvements in learning. The average student spent only 6 minutes on HIV-related pages. This potentially reconciles the findings in Derksen et al. (2022) with the null effects on HIV knowledge presented here.

¹⁸The relative frequency of each term across groups is measured using term frequency-inverse document frequency, a common heuristic for how important a word is within one set of responses relative to that word’s importance in the overall set of responses (Hvitfeldt and Silge, 2021).

Ruling out Alternative Explanations

It is unlikely that our findings are explained by the quality of Wikipedia as an information source, in terms accessibility or information available. First, students did not find Wikipedia to be difficult or unreliable. [Derksen et al. \(2022\)](#) show, using a number of measures, that the students were able to use Wikipedia to find information quickly and reliably, and preferred Wikipedia to other information sources. In the appendix we reproduce a table from [Derksen et al. \(2022\)](#) showing that at endline, students in the treatment group found information on Wikipedia to be easy to find and understand (Appendix Table [A11](#)). Second, the information needed to correctly answer the sexual health knowledge questions is readily available on Wikipedia, often on multiple pages that are of good quality. For example, as of January 2025, the answer to questions about the timing of a woman’s fertile window can be found on the pages for “Pregnancy Rate,” and “Pregnancy,” which are both B-class pages. The answer to the question about the HIV transmission rate can be found on the page “HIV/AIDS,” which has a GA-class rating, one of the highest quality ratings a page can have.

While information on health risks can be complex, we do not believe that our findings can be explained by a lack of understanding. If students did not understand the questions asked, we would expect to see variation in responses and high reported uncertainty. Instead, responses are tightly clustered around common misconceptions, and expressed with confidence. Nearly all students are misinformed on the topic of HIV risk; this implies that most students report incorrect beliefs in any large subsample of students, including the set of above-median achievers and the set of students attending competitive national secondary schools. Misconceptions persist for several years after graduation. When students are provided with correct information, most appear to adjust their beliefs.

Alternatively, perhaps students did search for and find the required information online, but discarded it on the advice of more trusted sources. Yet, students report Wikipedia to be highly trustworthy (Appendix Table [A11](#)), and used it for both incentivized quizzes and to prepare for high-stakes academic exams ([Derksen et al., 2022](#)). Conflicting information should, at the very least, lead to more variation in responses across students or a decrease in certainty over time, neither of which appear in our data. This explanation is also inconsistent with the sharp rise in beliefs we observe after providing brief informative statements to participants. It is more likely that most students did not search for this information at all; the average student spent only 6 minutes on Wikipedia pages related to HIV/AIDS.

It is important to note that our specific knowledge questions do not capture all important knowledge related to sexual health, and that online information may impact sex-related knowledge, attitudes and behavior in ways we cannot measure. For example, all students, and LGBT students in particular, may benefit from learning about sexual-

ity, and about topics that are taboo in Malawian society. More broadly, while internet access alone may not combat strongly-held misconceptions, it does impact general health knowledge among many dimensions ([Abbasi and Pongou, 2023](#)).

5 Conclusion

This study shows that providing adolescents in Malawi with access to Wikipedia, a high-quality online information source, does not counteract critical misinformation about HIV and pregnancy risks. Misconceptions persist over years, despite eventual access to the broader internet. This lack of impact appears to stem from students' certainty in their existing beliefs, leading to limited engagement with relevant health information, and from the breadth of other, more compelling, content available on platforms like Wikipedia. While the internet may offer various benefits, it does not serve as a useful substitute for targeted and effective sex education to address ingrained misconceptions about sexual health.

Limitations of the study include the potential for spillovers between treatment and control students, the fact that the sample size does not allow us to estimate subgroup effects with high statistical power, and that sexual behavior data is self-reported. These limitations are mitigated by the fact that the responses to the knowledge questions are near-universally and persistently incorrect. It is possible that student responses are biased by a desire to appear prudent. However, the answers given by nearly all students are not only incorrect but extreme. While self-reported sexual behavior data may also be subject to some desirability bias, approximately one-third of female students, and nine out of ten male students report unprotected sex in the past month. Self-reported HIV testing rates are well below optimal for young adults, who should be tested at least once per year according to Malawian national guidelines.

Our findings suggest that to become correctly informed about sexual health risks, young people need access to effective sex education that combats misinformation directly. Simple information interventions have proven highly effective in this regard ([Dupas, 2011a](#); [Angrist et al., 2019](#); [Derksen et al., 2022](#); [Kerwin, 2025](#)). The design and implementation of effective and scalable sex education programming, to reach not only students but also vulnerable out-of-school youth, remains an important subject for future research and policy evaluation. More broadly, future research should shed light on the causes of widespread and persistent health misinformation, as well as the potential for correct information to spread, impact beliefs, and affect health behavior.

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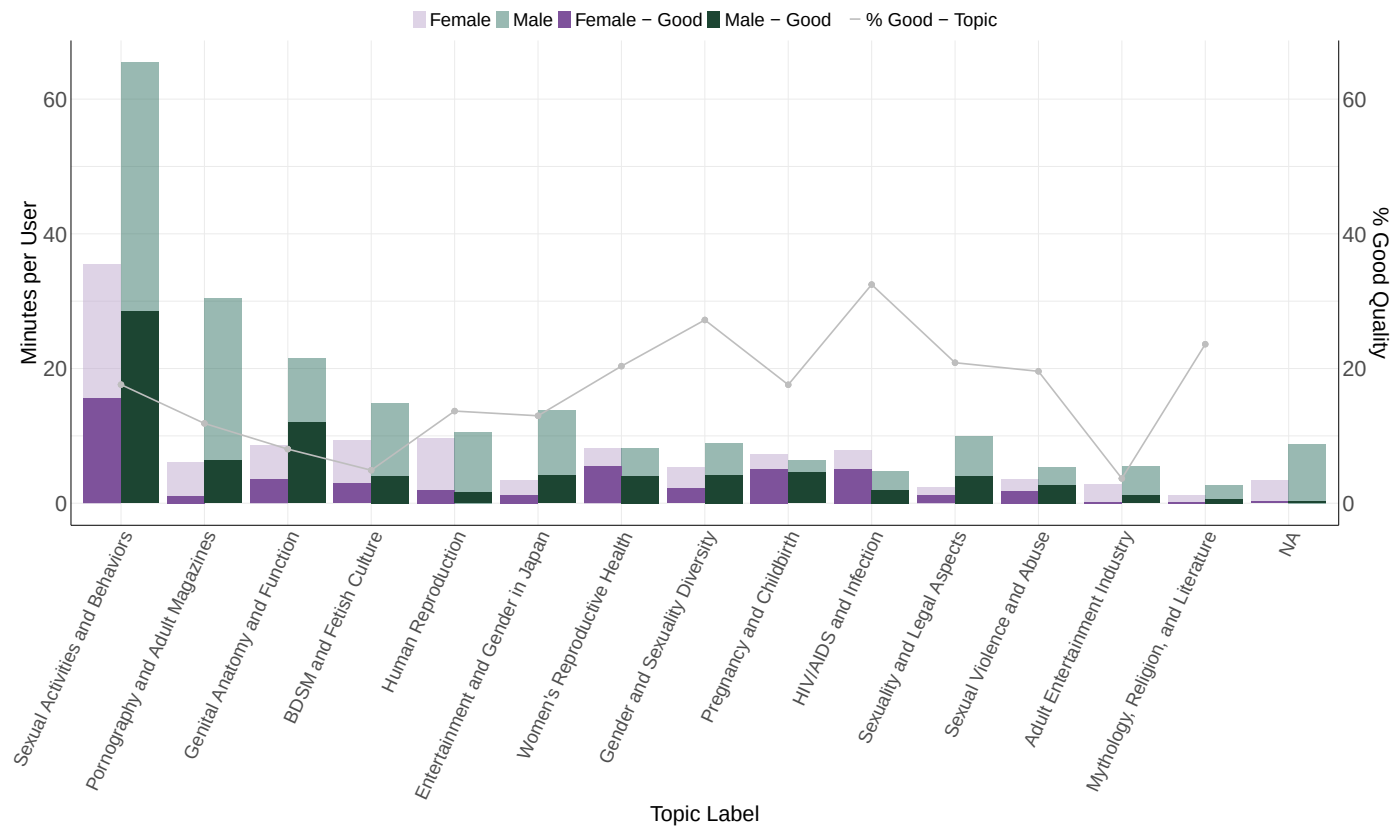
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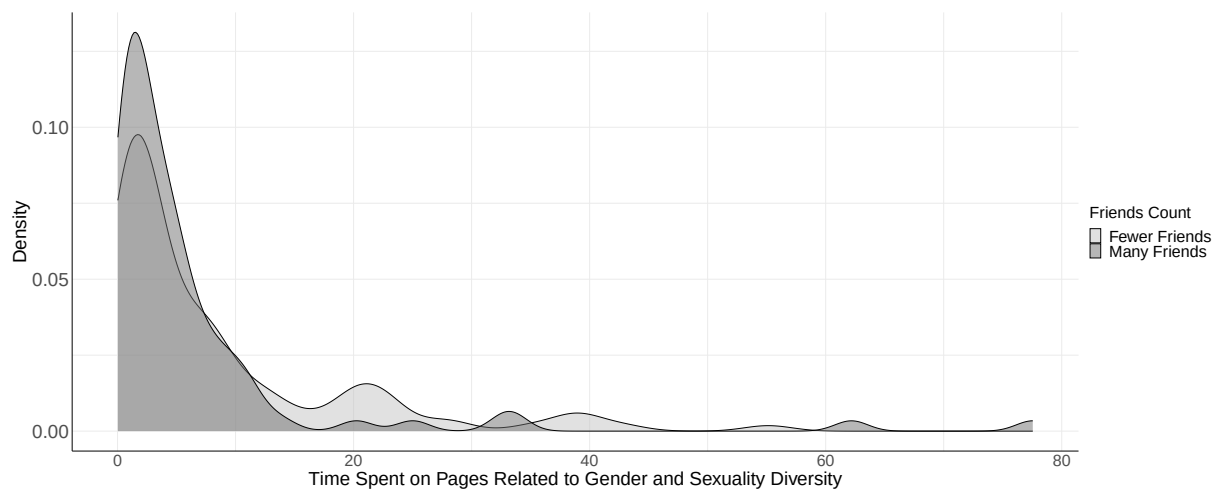
Figures

Figure 1: Browsing Time and Quality on Sex-Related Pages



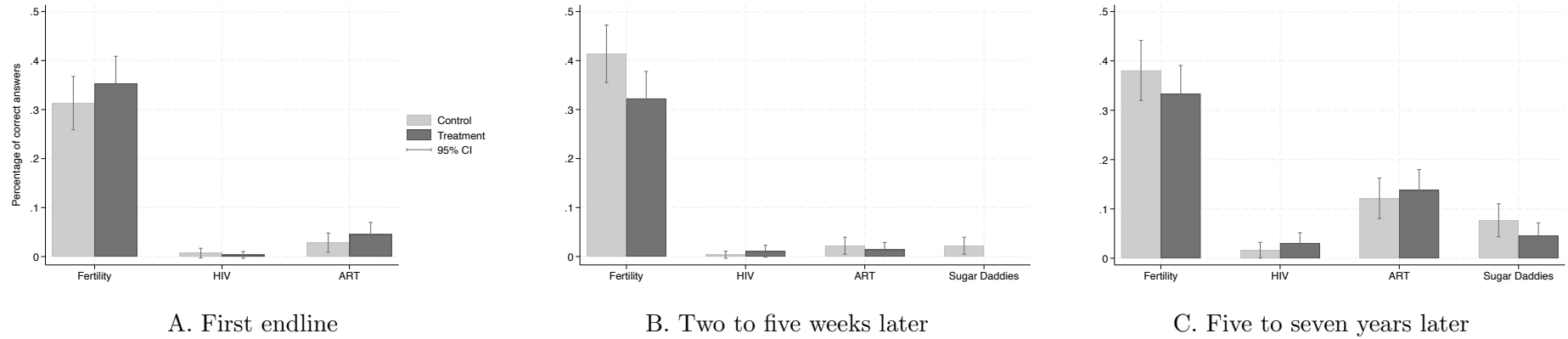
Notes: This figure shows the total amount of time students in the treatment group spend on pages related to different sexuality-related topics and the average quality of pages in each of these topics. Within each bar, the darker portion represents the amount of time students spent on good-quality pages in this topic. The page quality measure refers to pages that are assigned B-class or higher by Wikipedia editors, indicating the page is mostly complete and does not contain substantial errors. The solid grey line describes the average quality of pages in each respective topic. Topics are ranked from left to the right according the average number of minutes students spent on the topic by all students.

Figure 2: Distribution of Time Spent per User on Topics Related to Gender and Sexuality Diversity



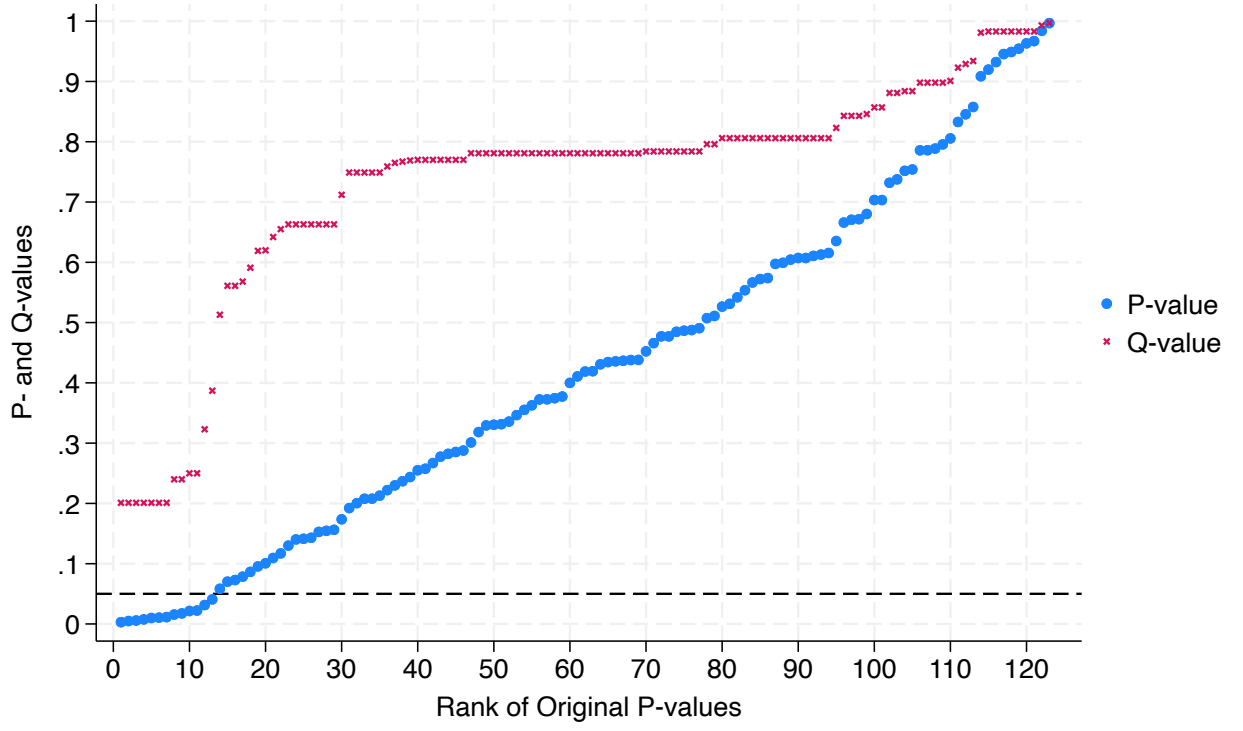
Notes: This figure shows the total amount of time students in the treatment group spent on Wikipedia pages related to “Gender and Sexuality Diversity,” according to our topic model. The darker curve shows the distribution of time spent on these pages by students who are more socially isolated (i.e., they have “fewer friends” compared to the median number of friends reported at the beginning of the intervention). The lighter curve shows the distribution for students who had more than the median number of friends at the beginning of the intervention.

Figure 3: Knowledge by Treatment Group



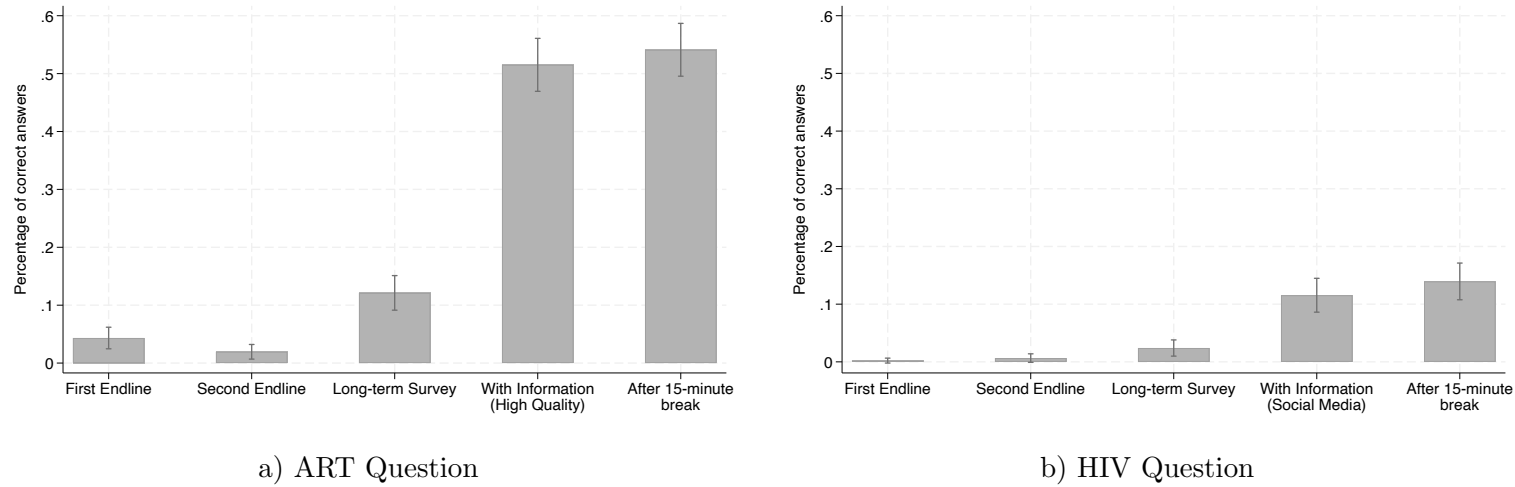
Notes: This figure shows the percentage of correct answers by treatment status at first endline (Panel A), two to five weeks after the first endline (Panel B), and five to seven years after the intervention (Panel C). The darker bar shows the percentage of correct answers for treated students and the lighter bar is for students in the control group. *Fertility* refers to the question about the timing of a woman's fertility window, *HIV* refers to the question about HIV transmission rates, *ART* refers to the question about the effectiveness of antiretroviral treatment in preventing HIV transmission, and *Sugar Daddies* refers to the question about the likelihood of HIV-infection for men by age. See Appendix Table A2 for the complete set of questions and their multiple-choice options. Confidence intervals at the 95 percent level are displayed.

Figure 4: P- and Q-values



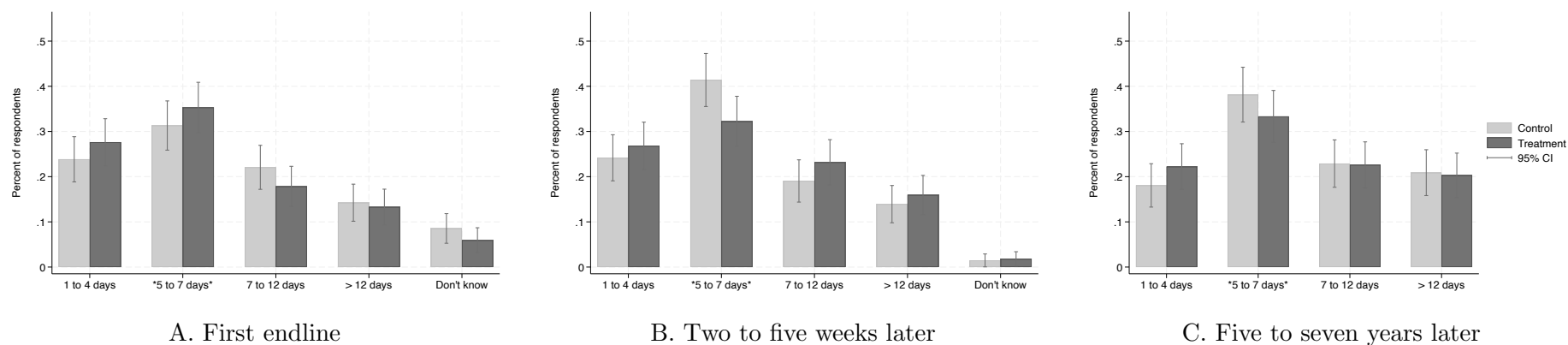
Notes: This graph plots p-values and false-discovery-rate (FDR) q-values, following [Anderson \(2008\)](#). These are calculated across the full set of estimates in the paper (i.e., Tables 2 to 7), including estimates of heterogeneous treatment effects.

Figure 5: Percentage of Correct Answers - ART and HIV Questions with Information



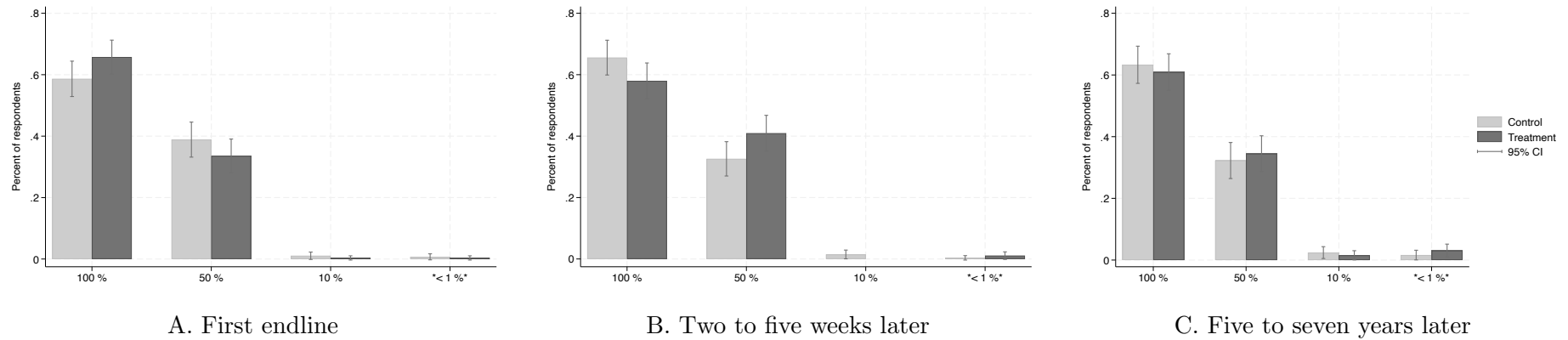
Notes: This figure shows the percentage of respondents who correctly answered the ART and HIV Questions. *Second Endline* took place two to five weeks after the *First Endline* in 2018. *Long-run Survey* is the percentage of correct answers in the long-run survey without any prompts or information. *With Information* is the percentage of correct answers after receiving information on the correct answer (from the United Nations for the ART Question and from an anonymous Twitter user for the HIV Question) and *After 15-minute Break* is their answers after taking a break in the long-run survey right after the *With Information* responses. See Appendix Table A2 for the complete set of questions and their multiple-choice options. The ART knowledge question is phrased slightly differently after the information provision to better match the United Nations statement: “Do you believe that it is impossible for a person with an undetectable viral load to spread HIV?”

Figure 6: Answers to Fertility Question by Treatment Group (Correct Answer: *5 to 7 days*)



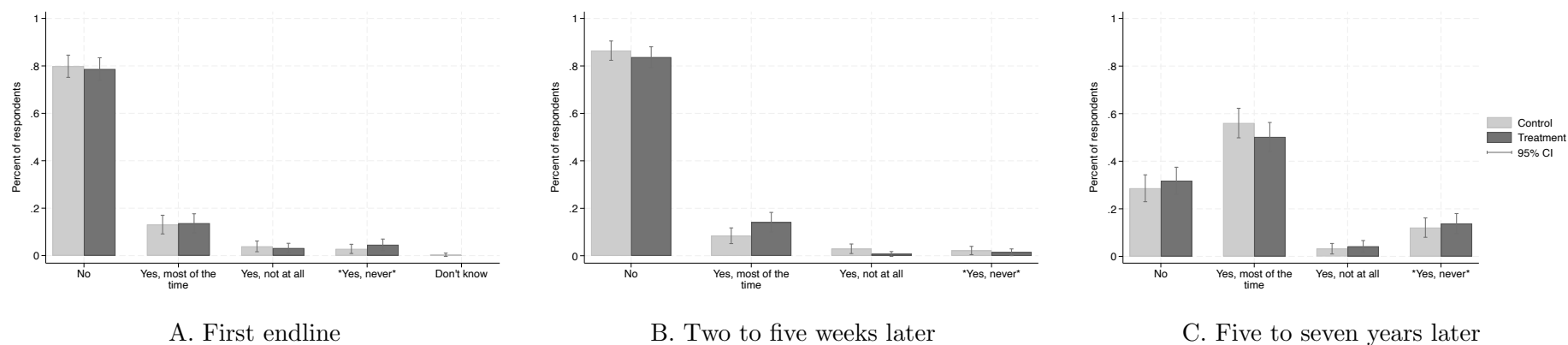
Notes: This figure shows the distribution of the answers by treatment status to the *Fertility* question on the timing of the fertile window for women at the first endline (Panel A), two to five weeks later (Panel B), and five to seven years later (Panel C). See Appendix Table A2 for the complete question and its multiple-choice options. The darker bar shows the percentage of correct answers for treated students and the lighter bar is for students in the control group. Confidence intervals at the 95 percent level are displayed.

Figure 7: Answers to HIV Question by Treatment Group (Correct Answer: $* < 1\% *$)



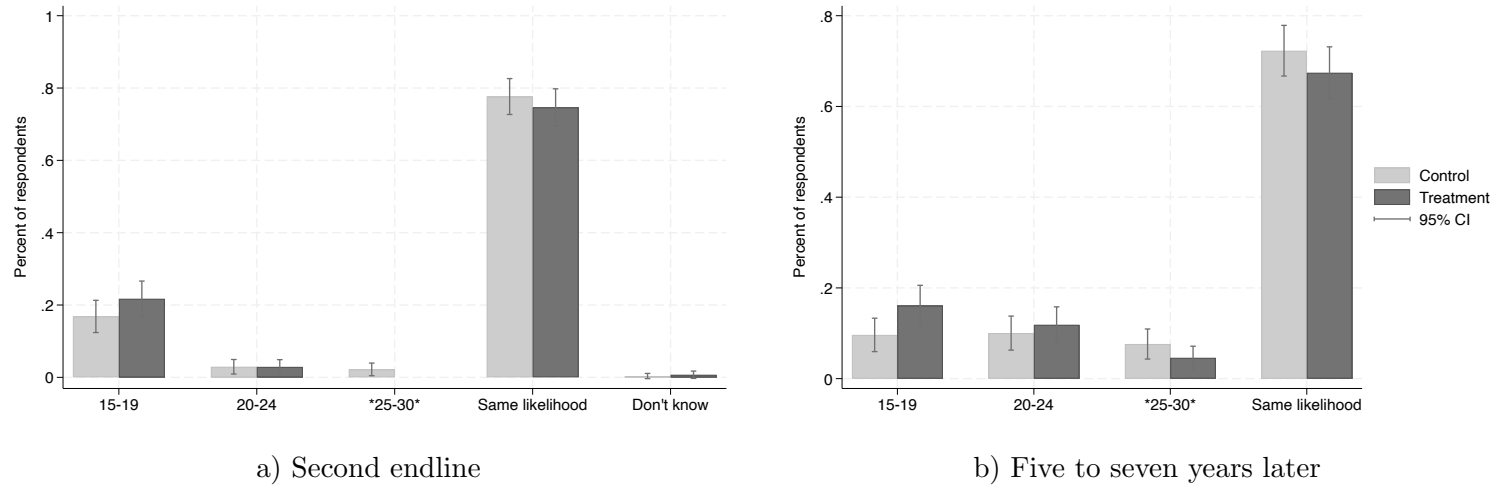
Notes: This figure shows the distribution of the answers by treatment status to the *HIV* question on the HIV transmission rate at the first endline (Panel A), two to five weeks later (Panel B), and five to seven years later (Panel C). See Appendix Table A2 for the complete question and its multiple-choice options. The darker bar shows the percentage of correct answers for treated students and the lighter bar is for students in the control group. Confidence intervals at the 95 percent level are displayed.

Figure 8: Answers to ART Question by Treatment Group (Correct Answer: **Yes, never**)



Notes: This figure shows the distribution of the answers to the *ART* question on the role of antiretroviral treatment in preventing HIV transmission at the first endline (Panel A), two to five weeks later (Panel B), and five to seven years later (Panel C). See Appendix Table A2 for the complete question and its multiple-choice options. The darker bar shows the percentage of correct answers for treated students and the lighter bar is for students in the control group. Confidence intervals at the 95 percent level are displayed.

Figure 9: Answers to “Sugar Daddies” Question by Treatment Group (Correct Answer: *25-30*)



Notes: This figure shows the distribution of the answers to the *Sugar Daddies* question on likelihood of HIV-infection for men by age at the second endline (Panel A) and five to seven years later (Panel B). See Appendix Table A2 for the complete question and its multiple-choice options. The darker bar shows the percentage of correct answers for treated students and the lighter bar is for students in the control group. Confidence intervals at the 95 percent level are displayed.

Tables

Table 1: Topic Model Coherence and Terms

Topic Label	Coherence	Prop. Good	Top 10 Terms
Adult Entertainment Industry	0.143	0.037	film, born, american, awards, adult, city, york, pornographic, award, films
BDSM and Fetish Culture	0.184	0.490	bds, bondage, fetish, play, fetishism, body, erotic, clothing, form, worn
Entertainment and Gender in Japan	0.032	0.130	women, japanese, male, term, dance, female, clubs, club, japan, strip
Gender and Sexuality Diversity	0.105	0.272	sexual, gender, people, gay, women, lgbt, sexuality, orientation, lesbian, social
Genital Anatomy and Function	0.104	0.080	penis, vagina, urethra, male, vulva, skin, blood, glans, labia, clitoris
HIV/AIDS and Infection	0.337	0.325	hiv, infection, disease, risk, aids, infections, symptoms, include, people, health
Human Reproduction	0.214	0.137	sperm, male, female, reproductive, human, semen, reproduction, species, egg, humans
Mythology, Religion, and Literature	0.064	0.236	character, series, love, greek, church, catholic, story, book, roman, god
Pornography and Adult Magazines	0.154	0.119	pornography, magazine, pornographic, erotic, published, adult, films, video, magazines, playboy
Pregnancy and Childbirth	0.233	0.176	pregnancy, birth, baby, medical, mother, women, childbirth, health, delivery, weeks
Sexual Activities and Behaviors	0.123	0.176	sexual, sex, activity, person, term, partner, intercourse, form, stimulation, orgasm
Sexual Violence and Abuse	0.137	0.196	rape, sexual, child, abuse, violence, victims, assault, united, women, victim
Sexuality and Legal Aspects	0.135	0.209	sex, prostitution, marriage, countries, legal, law, country, laws, women, united
Women’s Reproductive Health	0.161	0.204	vaginal, cancer, women, menstrual, vagina, pain, uterus, symptoms, breast, include

Notes: This table summarizes the top ten terms associated with each of the sexuality-related topics from the topic model, based on the introductions from the Wikipedia pages. These terms have been used to define the label for each topic. Coherence is a measure of how often words in a topic are used together. Higher levels of coherence indicate that the terms in that topic are more frequently associated with one another. “Prop. Good” reports the proportion of pages within this topic that meet the definition of good-quality pages, based on a page being classified as B-class or better in Wikipedia’s page classification system.

Table 2: Wikipedia Access and Health Knowledge

	First endline			2-5 weeks after first endline			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Fertility	HIV	ART	Fertility	HIV	ART	Sugar Daddies
Panel A: Average Treatment Effect							
Treatment	0.044 (0.040)	-0.005 (0.006)	0.018 (0.016)	-0.093** (0.039)	0.006 (0.007)	-0.006 (0.012)	-0.021** (0.009)
Panel B: Heterogeneous Treatment Effect							
Treatment $\times$ male	0.053 (0.054)	-0.006 (0.011)	-0.000 (0.018)	-0.065 (0.052)	0.006 (0.011)	-0.018 (0.015)	-0.009 (0.008)
Treatment $\times$ female	0.032 (0.061)	-0.004 (0.004)	0.039 (0.027)	-0.138** (0.060)	0.008 (0.009)	0.010 (0.018)	-0.038** (0.018)
Male	-0.097 (0.079)	0.024 (0.017)	-0.029 (0.032)	-0.164** (0.078)	0.012 (0.015)	0.031 (0.026)	-0.020 (0.015)
Mean control	0.313	0.007	0.028	0.414	0.004	0.022	0.022
Mean control - male	0.312	0.013	0.025	0.340	0.006	0.025	0.013
Mean control - female	0.315	0.000	0.032	0.518	0.000	0.018	0.035
Observations	567	569	569	549	549	549	549

Notes: This table shows the average and heterogeneous treatment effects of access to Wikipedia on knowledge around key sexual health misconceptions at first endline (Columns 1 to 3) and two to five weeks after the first endline (Columns 4 to 7). In each column, the outcome is an indicator variable that equals one if the answer is correct. *Fertility* refers to the question about the timing of a woman's fertility window, *HIV* refers to the question about HIV transmission rates, *ART* refers to the question about the effectiveness of antiretroviral treatment in preventing HIV transmission, and *Sugar Daddies* refers to the question about the likelihood of HIV-infection for men by age. See Appendix Table A2 for the complete set of questions and their multiple-choice options. Panel A shows the average treatment effect and Panel B presents the heterogeneous effects by gender. All regressions include a strata fixed effect. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Wikipedia Access, Knowledge and Misinformation Indices

	First endline		2-5 weeks after first endline	
	(1) Knowledge	(2) Misinformation	(3) Knowledge	(4) Misinformation
Panel A: Average Treatment Effect				
Treatment	0.029 (0.046)	-0.016 (0.051)	-0.099** (0.039)	0.040 (0.045)
Panel B: Heterogeneous Treatment Effect				
Treatment $\times$ male	0.005 (0.071)	0.023 (0.060)	-0.075 (0.047)	0.099* (0.055)
Treatment $\times$ female	0.058 (0.060)	-0.062 (0.086)	-0.130* (0.069)	-0.039 (0.075)
Male	0.015 (0.111)	0.063 (0.087)	-0.045 (0.078)	-0.105 (0.093)
Mean control	0.000	0.002	0.000	0.001
Mean control - male	0.025	0.030	-0.047	0.024
Mean control - female	-0.032	-0.033	0.066	-0.031
Observations	569	568	549	549

Notes: This table shows the average and heterogeneous treatment effects of access to Wikipedia on knowledge around key sexual health misconceptions at first endline (Columns 1 and 2) and two to five weeks after the first endline (Columns 3 and 4). The outcome in Columns 1 and 3 is a knowledge index constructed using indicator variables on whether the three or four health questions asked are answered correctly. See Appendix Table A2 for the complete set of questions and their multiple choice options. The outcome in Columns 2 and 4 is an index constructed from indicator variables that equal one if the answer to the question is incorrect and the respondent is certain about their incorrect answer. All indices are constructed following Anderson (2008). Panel A shows the average treatment effect and Panel B presents the heterogeneous effects by gender. All regressions include a strata fixed effect. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Wikipedia Access and Long-Run Health Knowledge

	Knowledge Questions				Indices	
	(1) Fertility	(2) HIV	(3) ART	(4) Sugar Daddies	(5) Knowledge	(6) Misinformation
Panel A: Average Treatment Effect						
Treatment	-0.051 (0.044)	0.012 (0.014)	0.024 (0.030)	-0.030 (0.022)	-0.033 (0.048)	0.032 (0.054)
Panel B: Heterogeneous Treatment Effect						
Treatment $\times$ male	-0.005 (0.059)	0.010 (0.023)	0.045 (0.037)	-0.046 (0.031)	-0.017 (0.067)	0.020 (0.073)
Treatment $\times$ female	-0.114* (0.068)	0.013 (0.013)	-0.002 (0.052)	-0.005 (0.027)	-0.055 (0.071)	0.043 (0.083)
Male	0.001 (0.102)	0.046* (0.024)	-0.046 (0.059)	-0.024 (0.042)	-0.008 (0.097)	0.081 (0.114)
Mean control	0.382	0.016	0.121	0.076	-0.000	0.001
Mean control - male	0.338	0.029	0.079	0.101	-0.007	0.074
Mean control - female	0.436	0.000	0.174	0.045	0.008	-0.092
Observations	510	509	509	510	510	510

Notes: This table shows the average and heterogeneous treatment effects of access to Wikipedia on long-run knowledge around key sexual health misconceptions, disaggregated by gender. In Columns 1 to 4, the outcome is an indicator variable that equals one if the answer is correct. *Fertility* refers to the question about the timing of a woman's fertility window, *HIV* refers to the question about HIV transmission rates, *ART* refers to the question about the effectiveness of antiretroviral treatment in preventing HIV transmission, and *Sugar Daddies* refers to the question about the likelihood of HIV-infection for men by age. See Appendix Table A2 for the complete set of questions and their multiple-choice options. Outcomes in Columns 5 and 6 are indices constructed following Anderson (2008). Column 5 is a knowledge index constructed using indicator variables on whether the three or four health questions asked are answered correctly. Column 6 is an index constructed from indicator variables that equal one if the answer to the question is incorrect and the respondent is certain about their incorrect answer. Panel A shows the average treatment effect and Panel B presents the heterogeneous effects by gender. All regressions include a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Wikipedia Access and Long-Run Sexual Health Behaviors

	Individual Outcomes					Indices	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sex before 18	Married before 20	Pregnancy before 20	Unprotected Sex	Tested for HIV	(1) to (4)	HIV Risk
Panel A: Average Treatment Effect							
Treatment	-0.079*** (0.028)	-0.003 (0.005)	-0.016 (0.011)	0.013 (0.046)	-0.030 (0.041)	-0.093** (0.043)	-0.038 (0.060)
Panel B: Heterogeneous Treatment Effect							
Treatment $\times$ male	-0.115*** (0.043)	-0.001 (0.003)	-0.026* (0.014)	-0.056 (0.063)	-0.054 (0.056)	-0.144** (0.056)	-0.134 (0.094)
Treatment $\times$ female	-0.040 (0.034)	-0.005 (0.015)	-0.002 (0.020)	0.046 (0.067)	-0.016 (0.064)	-0.033 (0.077)	0.060 (0.074)
Male	0.108 (0.067)	-0.032 (0.021)	-0.021 (0.027)	0.518*** (0.103)	0.282*** (0.094)	0.152 (0.124)	0.495*** (0.122)
Mean control	0.141	0.008	0.025	0.384	0.699	-0.002	-0.004
Mean control - male	0.198	0.000	0.023	0.482	0.732	0.058	0.177
Mean control - female	0.072	0.019	0.029	0.280	0.658	-0.078	-0.223
Observations	497	493	481	458	509	511	499

Notes: This table shows the average and heterogeneous treatment effects of access to Wikipedia on long-run sexual health behaviors, disaggregated by gender. The outcome in Column 1 is an indicator variable that equals one if the respondent declared having sex before 18 years old. In Columns 2 and 3, the outcomes are respectively indicator variables that equal one if the respondent was (2) married, or (3) had a pregnancy before 20 years old. In Column 4, the outcome is an indicator variable that equals one if the respondent declared having unprotected sex in the past month. In Column 5, the outcome is an indicator variable that equals one if the respondent declared testing for HIV in the past year. Outcomes in Columns 6 and 7 are indices constructed following [Anderson \(2008\)](#). Column 6 is a combination of all the outcomes from Columns 1 to 4. Column 7 is constructed using information on the total number of sexual partners in their life, the number of sexual partners in the past month, and an indicator variable that equals one if they had unprotected sex in the past month. Panel A shows the average treatment effect, and Panel B presents the heterogeneous effects by gender. All regressions include a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Wikipedia Access and Long-Run Attitudes towards Sex, Gender Diversity, and Norms

	Definition Index	Acceptance Indices			
	(1)	(2)	(3)	(4)	(5)
		Same-sex relationships	Transgender	Gender Norms	Sex Before Marriage
Panel A: Average Treatment Effect					
Treatment	-0.034 (0.067)	0.003 (0.056)	0.114 (0.070)	0.031 (0.050)	-0.286 (0.339)
Panel B: Heterogeneous Treatment Effect					
Treatment $\times$ male	-0.002 (0.095)	-0.074 (0.079)	-0.017 (0.094)	-0.046 (0.069)	-0.724 (0.478)
Treatment $\times$ female	-0.076 (0.096)	0.092 (0.081)	0.268*** (0.103)	0.138* (0.079)	0.359 (0.478)
Male	-0.026 (0.143)	0.246** (0.114)	0.383** (0.149)	-0.063 (0.122)	-0.753 (0.783)
Mean control	-0.001	0.000	-0.001	0.000	4.625
Mean control - male	-0.021	0.073	0.082	0.030	4.935
Mean control - female	0.024	-0.092	-0.105	-0.038	4.236
Observations	510	510	510	510	507

Notes: This table shows the average and heterogeneous treatment effects of access to Wikipedia on long-run attitudes towards sexuality and gender, disaggregated by gender. Outcomes in Columns 1 to 4 are indices constructed following [Anderson \(2008\)](#). The outcome in Column 1 uses categorical variables on whether the respondent can correctly define LGBT and transgender. See Appendix Table [A5](#) for examples of definitions provided by the respondents. The outcome in Column 2 include three distinct variables taking a value between 1 (never justified) and 10 (always justified) regarding homosexuality, homosexuality between two adult men, and two adult women, and two variables ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that homosexual activity should be not criminalized and whether same-sex marriage should be legalized. The outcome in Column 3 include two distinct variables taking a value between 1 (never justified) and 10 (always justified) regarding a person born male who lives and dresses like a woman and a person born female who lives and dresses like a man, and one variable ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that a person born male who lives like a woman should be free to dress as a woman at work without facing discrimination. The outcome in Column 4 includes one variable taking a value between 1 (never justified) and 10 (always justified) regarding abortion, and two variables ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that men make better executives than women (reversed score) and whether a husband should help his wife with household chores. See Appendix Figure [A6](#) for the distribution of these measures in the full sample. Finally, the outcome in Column 5 takes a value between 1 (never justified) and 10 (always justified) regarding sex before marriage. All regressions include a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Wikipedia Access and Long-Run Internet Use

Panel A: Access to a Smartphone			
	(1) Had Phone Before 20	(2) Minutes on Phone	(3) Has Digital Wellbeing
Treatment	0.042 (0.040)	4.293 (21.585)	-0.004 (0.044)
Treatment $\times$ male	0.024 (0.057)	-13.771 (34.162)	-0.086 (0.058)
Treatment $\times$ female	0.060 (0.055)	20.413 (26.524)	0.097 (0.069)
Male	-0.238*** (0.076)	-50.301 (42.499)	0.015 (0.092)
Mean control	0.604	271.761	0.440
Mean control - male	0.518	257.630	0.424
Mean control - female	0.712	289.246	0.459
Observations	511	270	511
Panel B: Internet Use			
	Days on Internet	Internet Bundles	Free Internet
Treatment	-0.063 (0.206)	-1161.289* (692.996)	0.060 (0.042)
Treatment $\times$ male	-0.319 (0.301)	-794.657 (1014.002)	0.008 (0.057)
Treatment $\times$ female	0.245 (0.264)	-1653.848* (939.253)	0.125* (0.065)
Male	-0.384 (0.444)	1606.329 (1259.527)	0.022 (0.078)
Mean control	5.560	7020.014	0.402
Mean control - male	5.439	7130.219	0.403
Mean control - female	5.712	6868.333	0.400
Observations	510	442	509
Panel C: Wikipedia Use			
	Used Wikipedia	Wikipedia Pages	Has Browsing History
Treatment	0.083*** (0.027)	0.089 (0.814)	-0.016 (0.041)
Treatment $\times$ male	0.061* (0.034)	0.949 (0.673)	-0.031 (0.058)
Treatment $\times$ female	0.113*** (0.042)	-0.541 (1.319)	-0.003 (0.054)
Male	0.067 (0.059)	-3.907*** (1.079)	-0.323*** (0.080)
Mean control	0.855	5.130	0.644
Mean control - male	0.885	2.486	0.518
Mean control - female	0.818	7.270	0.802
Observations	510	325	511
Panel D: Social Media, Wikipedia, and Health			
	Social Media Usage	Social Media and Health	Wikipedia and Health
Treatment	0.003 (0.036)	-0.053 (0.052)	0.076 (0.066)
Treatment $\times$ male	-0.033 (0.047)	-0.032 (0.068)	0.115 (0.090)
Treatment $\times$ female	0.050 (0.057)	-0.081 (0.081)	0.018 (0.096)
Male	0.085 (0.071)	-0.103 (0.106)	-0.466*** (0.132)
Mean control	0.021	-0.000	-0.001
Mean control - male	-0.007	-0.035	-0.132
Mean control - female	0.057	0.043	0.165
Observations	510	510	510

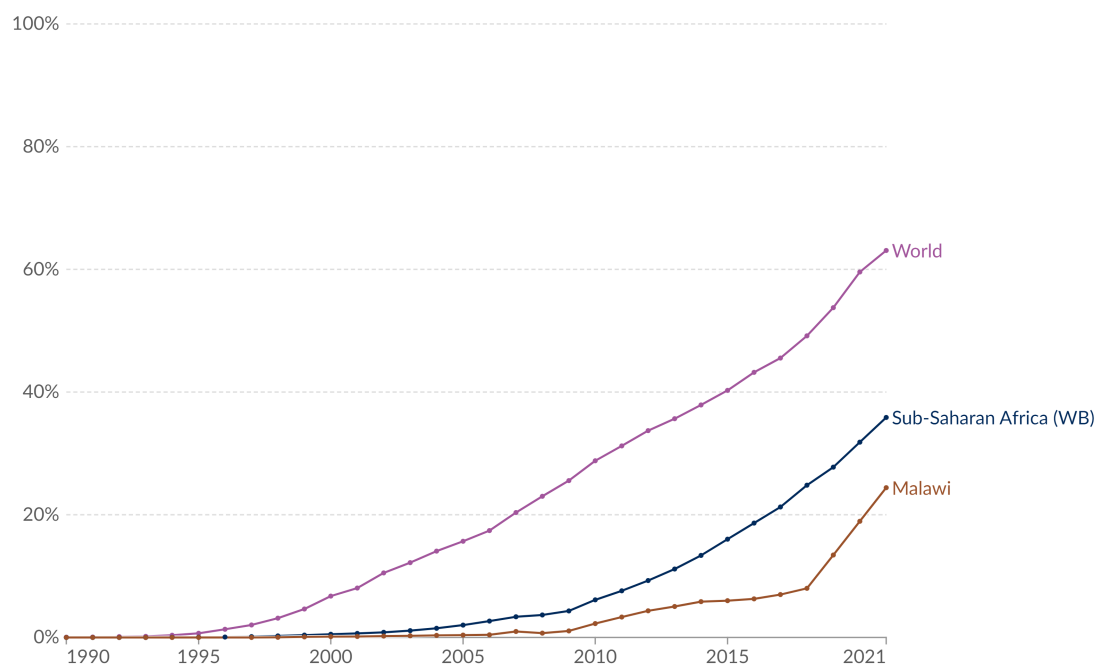
Notes: This table shows the average and heterogeneous treatment effect of access to Wikipedia on long-run internet use, disaggregated by gender. Panel A. Access to smartphone: (1) Owned phone before 20 years old; (2) Minutes on smartphone last Wednesday/Saturday; (3) Has Digital Wellbeing Panel B. Internet usage: (1) Days they used the internet over the past week; (2) Amount spent in Malawian Kwacha on internet bundles; (3) Accessed internet for free in the past month. Panel C. Wikipedia usage: (1) Ever used Wikipedia; (2) Number of Wikipedia pages they looked at in the past week; (3) Have browsing history. Panel D. Social media usage and online information on health. All indices are constructed following [Anderson \(2008\)](#): (1) Amount spent on social media bundles, minutes using social media last Wednesday/Saturday, how often they used social media in the past month, and have social media accounts (Facebook, Twitter, Tiktok, and WhatsApp); (2) Use social media for information about health and social media in top 3 best place to find information about health; (3) Use Wikipedia for information about health and Wikipedia in top 3 best place to find information about health. Outcomes in Columns 3 from Panels A, B, and C were not pre-specified in the pre-analysis plan. All regressions include a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Figures

Appendix Figure A1: Share of Individuals Using the Internet

Share of the population using the Internet

Share of the population who used the Internet¹ in the last three months.



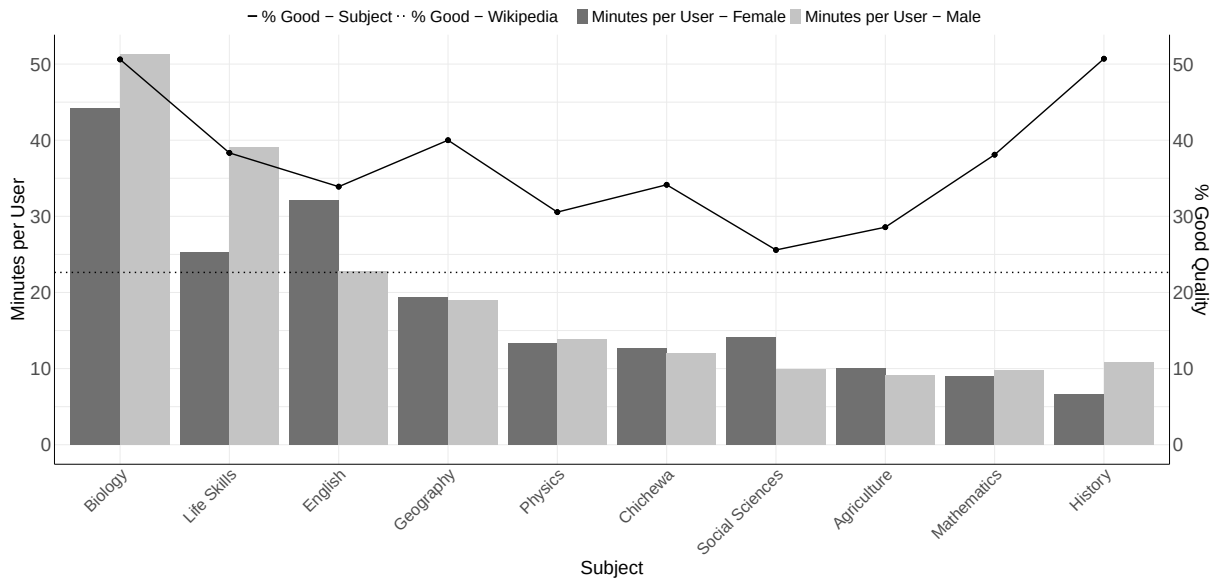
Data source: International Telecommunication Union (via World Bank)

OurWorldinData.org/internet | CC BY

1. **Internet user:** An internet user is defined by the International Telecommunication Union as anyone who has accessed the internet from any location in the last three months. This can be from any type of device, including a computer, mobile phone, personal digital assistant, games machine, digital TV, and other technological devices.

Notes: Figure reproduced from <https://ourworldindata.org/internet> (Ritchie et al., 2023), accessed on 16 October, 2024.

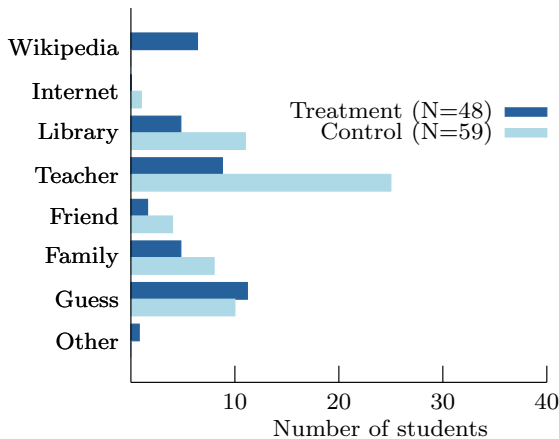
Appendix Figure A2: Browsing Time and Quality on Syllabus-Related Pages



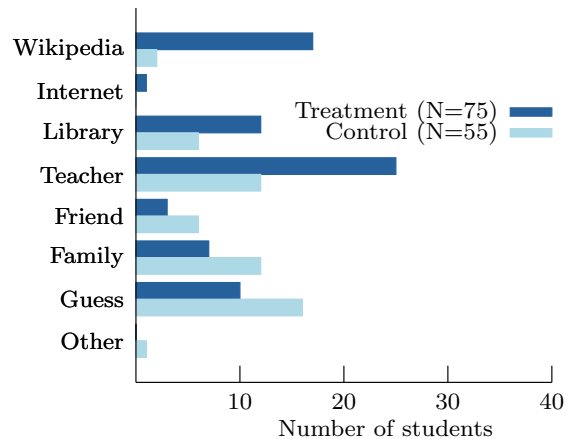
Notes: This figure describes the total amount of time students spend on pages related to the school syllabus by gender. The solid black line describes the average quality of pages in each respective subject. The black horizontal dash line corresponds with the average page quality of all pages that students searched during the experiment.

Appendix Figure A3: Source of Information: Female Students on the Fertility Question

Panel A. Correct Answer

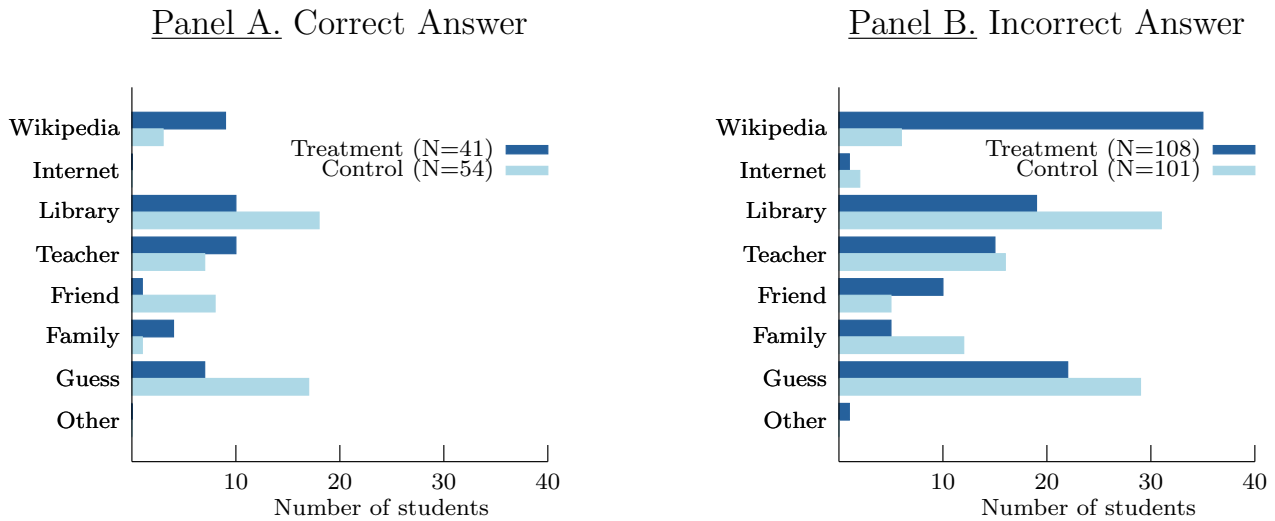


Panel B. Incorrect Answer



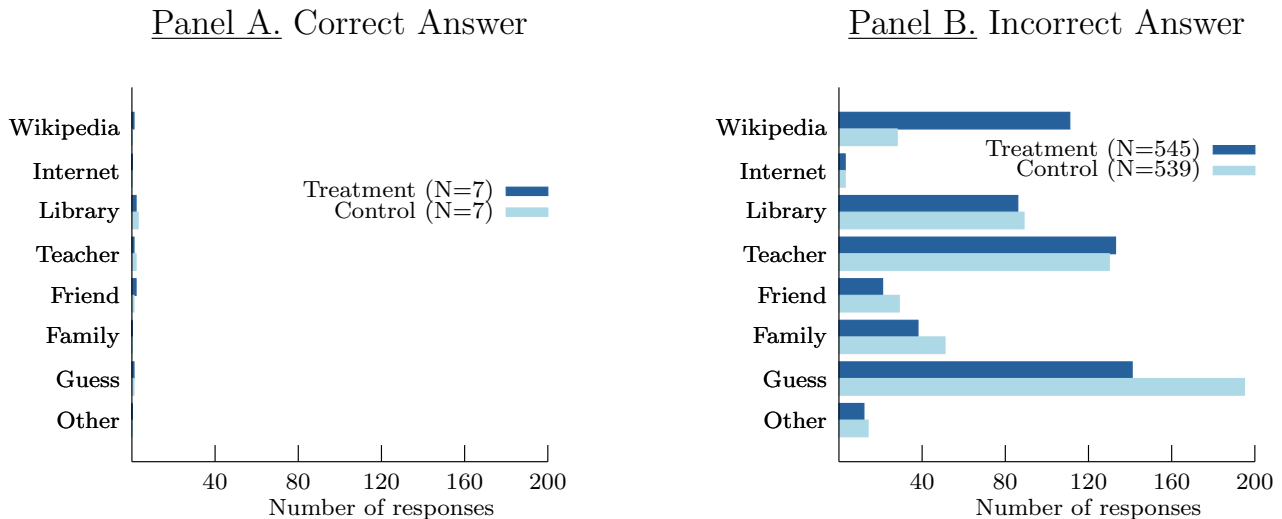
Notes: This figure shows the source of information to the correct (Panel A) and incorrect (Panel B) answers to the *Fertility* question on the timing of the fertile window for women by treatment status. See Appendix Table A2 for the complete question and its multiple-choice options. The sample only include female students. Data is collected at the second endline.

Appendix Figure A4: Source of Information: Male Students on the Fertility Question



Notes: This figure shows the source of information to the correct (Panel A) and incorrect (Panel B) answers to the *Fertility* question on the timing of the fertile window for women by treatment status. See Appendix Table A2 for the complete question and its multiple-choice options. The sample only include male students. Data is collected at the second endline.

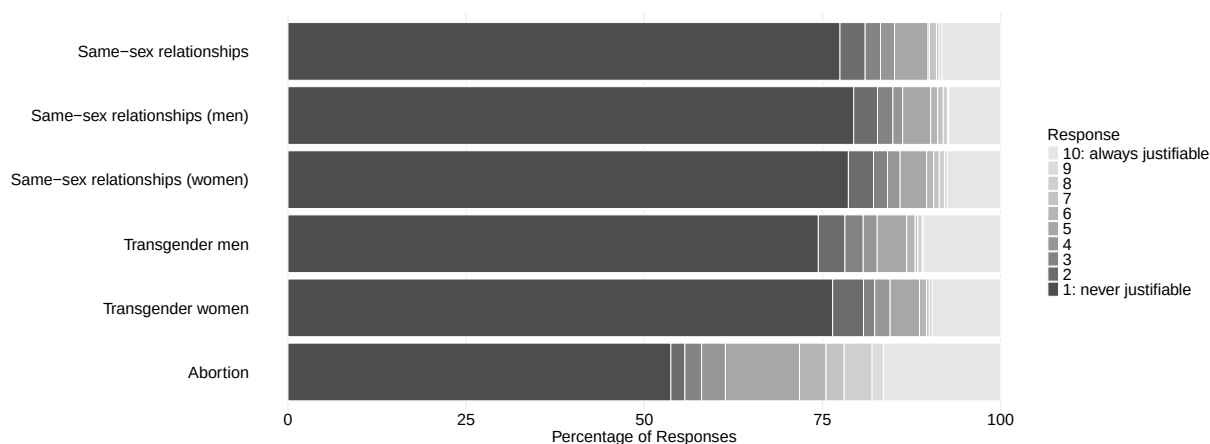
Appendix Figure A5: Source of Information: All Students on the HIV Questions



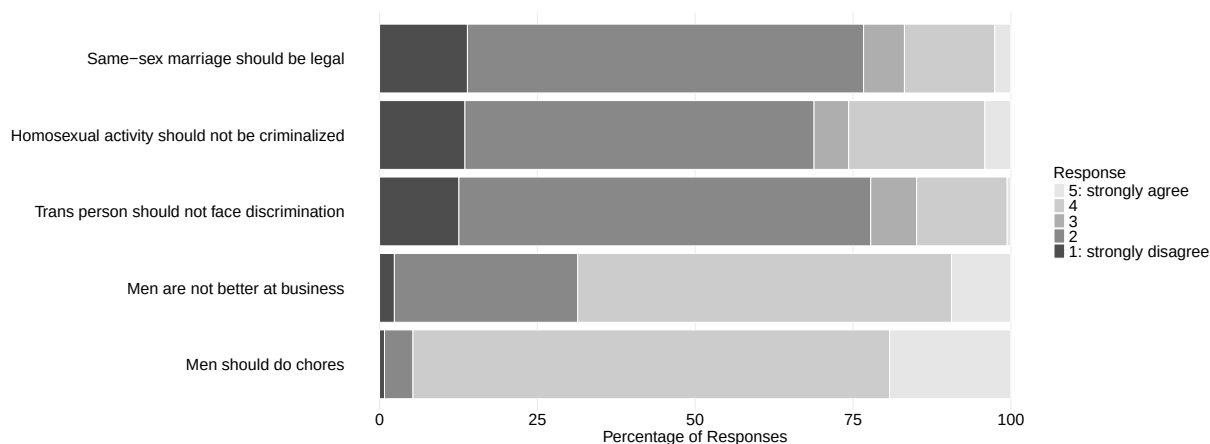
Notes: This figure shows the source of information to the correct (Panel A) and incorrect (Panel B) answers to the *HIV*, *ART*, and *Sugar Daddies* questions by treatment status. *HIV* refers to the question about HIV transmission rates, *ART* refers to the question about the effectiveness of antiretroviral treatment in preventing HIV transmission, and *Sugar Daddies* refers to the question about the likelihood of HIV-infection for men by age. See Appendix Table A2 for the complete set of questions and their multiple-choice options. Data is collected at the second endline.

Appendix Figure A6: Attitudes towards Gender Norms and Rights

Panel A. Justifiability scale (1-10)



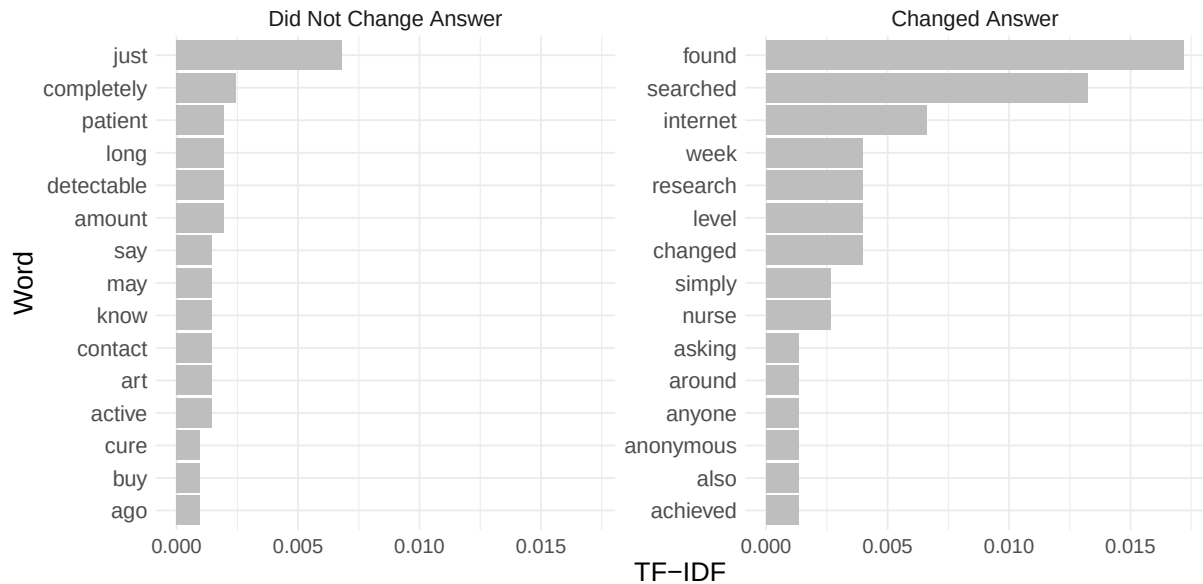
Panel B. Agreement level (1-5)



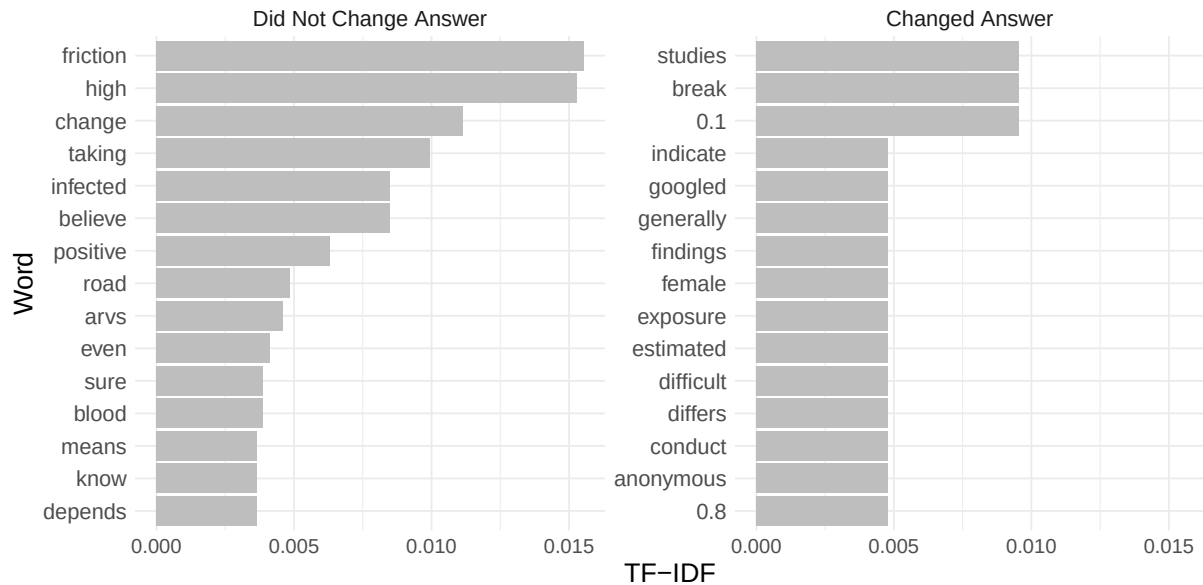
Notes: This figure describes the proportion of people reporting attitudes towards gender and sexuality-related norms and rights. Panel A includes six questions about how justifiable concepts were on a scale of 1 (never justifiable) to 10 (always justifiable). They were asked whether same-sex relationships (in general), same-sex relationships (male or female), transgender men, transgender women, and abortion were ever justifiable. The darkest bars indicate the proportion of people who responded an activity was never justifiable, the lightest bar indicates the proportion who responded an activity was always justifiable. Panel B included five questions about how strongly they agree with different statements. They were asked how strongly they agreed with the idea that same-sex marriage should be legal, homosexual activity should not be criminalized, trans people should not face discrimination, men are not better at business, and men should do chores. The darkest bars indicate the proportion who strongly disagreed, the lightest bars indicate the proportion who strongly agreed. The direction of the questions “homosexual activity should not be criminalized” and “men are not better at business” have been reversed to keep the interpretation consistent with other questions. The questions “men are not better at business” and “men should do chores” did not include a neutral option.

Appendix Figure A7: Descriptions of Answer Changes

Panel A. TF-IDF for Explanations of Response Changes - ART Question



Panel B. TF-IDF for Explanations of Response Changes - HIV Question



Notes: This figure describes the relative frequency of terms used in response to the question “Did you change your answer to this question? Why or why not?” after answering the ART Question (Panel A) or the HIV Question (Panel B) incorrectly before a 15-minute break in the long-run survey. These questions are described in Figure 5. The relative frequency of terms used in these responses is measured by each word’s “term frequency-inverse document frequency” (TF-IDF). Higher TF-IDF values indicate a term is relatively more important within that subset of responses. The figures on the left side of each panel show the the most important words for participants who did not change their incorrect response after a short break, the right side of each panel shows the most important words for participants who changed their response to the correct answer after a short break.

Appendix Tables

Appendix Table A1: Balance and Attrition in Survey Sample

	(1)	(2)	(3)
	Treatment	Control	p-value
Panel A. Balance			
Average exam score in English	57.188 (13.187)	57.626 (13.533)	0.694
Average examen score in Biology	53.810 (17.196)	53.267 (17.640)	0.709
Age	15.973 (1.971)	16.060 (1.845)	0.577
Female	0.452 (0.499)	0.430 (0.496)	0.583
District of origin	0.605 (0.490)	0.574 (0.495)	0.444
Mother's education	0.746 (0.436)	0.698 (0.460)	0.224
Father's education	0.849 (0.359)	0.852 (0.356)	0.918
Household has electricity	0.611 (0.488)	0.557 (0.498)	0.179
Household has a mobile phone	0.870 (0.336)	0.849 (0.359)	0.451
Panel B. Attrition			
First Endline	0.047 (0.211)	0.050 (0.219)	0.653
Second Endline	0.083 (0.276)	0.084 (0.278)	0.933
Long-term Survey	0.133 (0.340)	0.161 (0.368)	0.362
Number of students	301	298	

Notes: From [Derksen et al. \(2022\)](#). Panel A: Balance table across the treatment (N=301) and control (N=298) groups in the survey sample. Column 3 shows the p-value of the difference between treatment and control groups. District of origin equals 1 if the district where the student is from is the same district as the school district. Mother's and father's education is equal to one if the parent has completed primary education. Panel B: Attrition rates in the treatment and control groups. Estimates are the coefficient from a regression of attrition indicator in first and second endline surveys and the long-run survey (five to seven years later) on the treatment status, with strata fixed effects. Standard errors in parentheses.

Appendix Table A2: Questions and Multiple-Choice Options on Health Information Questions

Label	Question	Multiple-choice options
Fertility	Imagine a woman who has a menstrual cycle every month. She will be fertile on certain days every month. This is called the fertile window. That means if she has sex on those days, she might get pregnant. How many fertile days does an average woman have every cycle?	(a) 1-4 days (b) 5-7 days (c) 7-12 days (d) More than 12 days (e) Don't know (f) Refuse to answer
HIV	I want to ask you about the HIV transmission rate. Imagine an HIV-positive man who is NOT on treatment (ARVs) has unprotected sex with an HIV-negative woman one time. What is the approximate chance that she gets HIV?	(a) 100% chance – she will definitely get HIV (b) 50% chance – she has a 50/50 chance of getting HIV (c) 10% chance – in a single time she will probably not get HIV (d) Less than 1% chance – in a single time she will probably not get HIV (e) Don't know (f) Refuse to answer
ART	Do you think ARVs can stop a person living with HIV from spreading the virus?	(a) No (b) Yes, ARVs can reduce the chance of spreading the virus but most of the time the virus will still spread (c) Yes, as soon as a person starts taking ARVs they cannot spread the virus at all (d) Yes, a person who has been taking ARVs properly for a long time will almost never spread the virus (e) Don't know (f) Refuse to answer
Sugar Daddies	Who is the most likely to be HIV-positive?	(a) A teenage boy aged 15-19 (b) A man aged 20-24 (c) A man aged 25-29 (d) They all have the same likelihood (e) Don't know (f) Refuse to answer

Notes: This tables presents the four questions asked at the two endlines and in the long-run follow-up survey, the multiple choice options that the respondent could select, and the correct answer highlighted in bold.

Appendix Table A3: Certainty about Health Information Answers

	Correct Answer				Incorrect Answer			
	Treatment		Control		Treatment		Control	
	(1) % Certain	(2) N	(3) % Certain	(4) N	(5) % Certain	(6) N	(7) % Certain	(8) N
Panel A: First Endline								
Fertility (female students)	0.936	47	0.923	39	0.771	83	0.815	81
Fertility (male students)	0.963	54	0.980	49	0.909	88	0.884	95
HIV questions (all students)	0.786	14	1.000	10	0.966	558	0.960	554
Panel B: Two to Five Weeks Later								
Fertility (female students)	0.958	48	0.966	59	0.868	76	0.889	54
Fertility (male students)	1.000	41	0.981	54	0.954	108	0.990	102
HIV questions (all students)	0.714	7	0.923	13	0.956	821	0.963	806
Panel C: Long-run Survey								
Fertility (female students)	0.564	39	0.562	48	0.679	78	0.613	62
Fertility (male students)	0.771	48	0.809	47	0.783	92	0.739	92
HIV questions (all students)	0.786	56	0.811	53	0.876	727	0.878	690
Panel D: Long-run Survey - Specific HIV Questions								
ART question	0.833	36	0.867	30	0.880	225	0.867	218
HIV question	0.875	8	0.750	4	0.889	253	0.881	243
Panel E: With Information								
ART* question	0.854	144	0.802	126	0.752	117	0.732	123
HIV question	0.914	35	0.897	29	0.881	226	0.891	220
Panel F: After 15-minute break								
ART* question	0.887	150	0.879	132	0.856	111	0.812	117
HIV question	0.976	41	0.971	34	0.935	216	0.902	215

Notes: This table displays the percentage and number of respondents who are certain about their correct (Columns 1 to 4) and incorrect (Columns 5 to 8) answers by treatment status. Panel A presents the statistics at the first endline and the HIV questions include the *HIV* and *ART* questions on the HIV transmission rate and the role of antiretroviral treatment in preventing HIV transmission. In Panels B and C, two to five weeks and five to seven years after the first endline, respectively, the HIV questions also include the *Sugar Daddies* question about the likelihood of HIV-infection for men by age. The *Fertility* question, displayed by gender, is on the timing of the fertile window for women. Panels D, E and F focus on the ART and HIV questions in the long-run survey. Panels D and E are asked later in the long-run survey, after receiving information about the correct answer (Panel D) and following a 15-minute break (Panel F). See Appendix Table A2 for the complete set of questions and their multiple-choice options. ART* is a modified version of the ART question: “Do you believe that it is impossible for a person with an undetectable viral load to spread HIV?” Certainty is constructed using information on whether the respondent is certain or very certain about their answer.

Appendix Table A4: Wikipedia Access and Long-Run Sexual Health Behaviors by School Type

	Individual Outcomes					Indices	
	(1) Sex before 18	(2) Married before 20	(3) Pregnancy before 20	(4) Unprotected Sex	(5) Tested for HIV	(6) (1) to (4)	(7) HIV Risk
Panel A: Average Treatment Effect							
Treatment	-0.079*** (0.028)	-0.003 (0.005)	-0.016 (0.011)	0.013 (0.046)	-0.030 (0.041)	-0.093** (0.043)	-0.038 (0.060)
Panel B: Heterogeneous Treatment Effect							
Treatment $\times$ co-educational	-0.091** (0.046)	0.001 (0.009)	-0.031 (0.020)	-0.063 (0.073)	-0.092 (0.060)	-0.151** (0.073)	-0.162 (0.102)
Treatment $\times$ sex-seggregated	-0.069* (0.035)	-0.007 (0.006)	-0.003 (0.012)	0.069 (0.059)	0.021 (0.058)	-0.043 (0.050)	0.067 (0.070)
Co-educational school	-0.007 (0.070)	-0.001 (0.006)	0.031 (0.034)	0.394* (0.211)	-0.153 (0.191)	0.233** (0.100)	0.376* (0.212)
Mean control	0.141	0.008	0.025	0.384	0.699	-0.002	-0.004
Mean control - male	0.198	0.000	0.023	0.482	0.732	0.058	0.177
Mean control - female	0.072	0.019	0.029	0.280	0.658	-0.078	-0.223
Observations	497	493	481	458	509	511	499

Notes: This table shows the average and heterogeneous treatment effect of access to Wikipedia on long-run sexual health behaviours, disaggregated by school type. See Table 5 for a description of the outcomes. Panel A shows the average treatment effect and Panel B presents the heterogeneous effects by school type. Two schools were co-educational and two schools were sex-seggregated. All regressions include a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A5: Examples of Definitions of a Transgender Person

Scores	Examples of Definitions
A. Related to gender identity	
3	An individual whose gender identity differs from sex
3	Yes.a transgender person is someone whose gender identity differs from that typically associated with sex they were assigned at birth.
3	It's when someone is being identified as the opposite gender, for example a Male being identified as a female
B. Related to a change in gender or sex	
2	Transgender is a person who changes their gender
2	It means changing their gender from either male to female or female to male
2	A transgender is when a person changes him/herself into the opposite sex. Let's say a person was born male but changes himself into female
C. Related to a change using surgery or hormones	
1	When a person undergo surgery to change his or sex from male to female or from female to male
1	Being a transgender means you were born a Male and then you do surgery to change your sex
1	To be a transgender it simply means that when let's say a boy having girl hormones or vise versa
D. Related to behavior	
1	A man behave like woman, and a woman behave like a man
1	It means that a person is let's say a male, can be acting like a female
1	A person who was born male but live life as a woman
E. Other examples	
0	A person who is attracted to both genders
0	Those people who have both reproductive system
0	A transgender is a person who practices homosexuality

Notes: Examples of definitions provided by respondents to the question “Do you know what it means for a person to be transgender?” We grouped all definitions in five main categories and gave a score out of three to each category. The score is used to construct the Definition Index in Column 1 of Table 6.

Appendix Table A6: Correlation between Social Isolation and LGBT Acceptance

	Acceptance Index	
	(1) Same-sex relationships	(2) Transgender
Fewer friends	0.030 (0.057)	-0.069 (0.072)
Observations	510	510

Notes: Correlation between network and acceptance towards gender norms and rights. Outcomes are indices constructed following [Anderson \(2008\)](#). The outcome in Column 1 includes three distinct variables taking a value between 1 (never justified) and 10 (always justified) regarding homosexuality, homosexuality between two adult men, and two adult women, and two variables ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that homosexual activity should be not criminalized and whether same-sex marriage should be legalized. The outcome in Column 2 includes two distinct variables taking a value between 1 (never justified) and 10 (always justified) regarding a person born male who lives and dresses like a woman and a person born female who lives and dresses like a man, and one variable ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that a person born male who lives like a woman should be free to dress as a woman at work without facing discrimination. Fewer friends is a dummy variable that equals one if the respondent had less friends than the median student at baseline. The correlation is constructed from a regression that includes a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A7: Wikipedia Access and Long-Run Health Knowledge, Excluding Time Trend

	Knowledge Questions				Indices	
	(1) Fertility	(2) HIV	(3) ART	(4) Sugar Daddies	(5) Knowledge	(6) Misinformation
Panel A: Average Treatment Effect						
Treatment	-0.049 (0.044)	0.012 (0.013)	0.022 (0.030)	-0.033 (0.022)	-0.038 (0.048)	0.027 (0.053)
Panel B: Heterogeneous Treatment Effect						
Treatment $\times$ male	-0.006 (0.059)	0.010 (0.023)	0.045 (0.037)	-0.045 (0.032)	-0.016 (0.067)	0.020 (0.073)
Treatment $\times$ female	-0.103 (0.067)	0.015 (0.013)	-0.006 (0.051)	-0.017 (0.027)	-0.065 (0.069)	0.038 (0.080)
Male	-0.050 (0.086)	0.034* (0.017)	-0.027 (0.046)	0.033 (0.040)	0.037 (0.080)	0.107 (0.095)
Mean control	0.382	0.016	0.121	0.076	-0.000	0.001
Mean control - male	0.338	0.029	0.079	0.101	-0.007	0.074
Mean control - female	0.436	0.000	0.174	0.045	0.008	-0.092
Observations	510	509	509	510	510	510

Notes: This table shows the average and heterogeneous treatment effect of access to Wikipedia on long-run knowledge around key sexual health misconceptions, disaggregated by gender, without the time trend. In Columns 1 to 4, the outcome is an indicator variable that equals one if the answer is correct. *Fertility* refers to the question about the timing of a woman's fertility window, *HIV* refers to the question about HIV transmission rates, *ART* refers to the question about the effectiveness of antiretroviral treatment in preventing HIV transmission, and *Sugar Daddies* refers to the question about the likelihood of HIV-infection for men by age. See Appendix Table A2 for the complete set of questions and their multiple-choice options. Outcomes in Columns 5 and 6 are indices constructed following Anderson (2008). Column 5 is a knowledge index constructed using indicator variables on whether the three or four health questions asked are answered correctly. Column 6 is an index constructed from indicator variables that equal one if the answer to the question is incorrect and the respondent is certain about their incorrect answer. Panel A shows the average treatment effect and Panel B presents the heterogeneous effects by gender. All regressions include a strata fixed effect. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A8: Wikipedia Access and Long-Run Sexual Health Behaviors, Excluding Time Trend

	Individual Outcomes					Indices	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Sex before 18	Married before 20	Pregnancy before 20	Unprotected Sex	Tested for HIV	(1) to (4)	HIV Risk
Panel A: Average Treatment Effect							
Treatment	-0.078*** (0.028)	-0.003 (0.005)	-0.016 (0.011)	0.006 (0.046)	-0.032 (0.041)	-0.095** (0.043)	-0.058 (0.061)
Panel B: Heterogeneous Treatment Effect							
Treatment $\times$ male	-0.116*** (0.043)	-0.001 (0.003)	-0.025* (0.014)	-0.056 (0.063)	-0.055 (0.056)	-0.144** (0.056)	-0.132 (0.093)
Treatment $\times$ female	-0.033 (0.032)	-0.006 (0.014)	-0.005 (0.019)	0.063 (0.065)	0.000 (0.062)	-0.032 (0.073)	0.041 (0.072)
Male	0.076 (0.058)	-0.026 (0.017)	-0.008 (0.025)	0.420*** (0.091)	0.201** (0.079)	0.146 (0.104)	0.591*** (0.118)
Mean control	0.141	0.008	0.025	0.384	0.699	-0.002	-0.004
Mean control - male	0.198	0.000	0.023	0.482	0.732	0.058	0.177
Mean control - female	0.072	0.019	0.029	0.280	0.658	-0.078	-0.223
Observations	497	493	481	458	509	511	499

Notes: This table shows the average and heterogeneous treatment effect of access to Wikipedia on long-run sexual health behaviors disaggregated by gender, without the time trend. The outcome in Column 1 is an indicator variable that equals one if the respondent declared having sex before 18 years old. In Columns 2 and 3, the outcomes are respectively indicator variables that equal one if the respondent was (2) married or (3) had a pregnancy before 20 years old. In Column 4, the outcome is an indicator variable that equals one if the respondent declared having unprotected sex in the past month. In Column 5, the outcome is an indicator variable that equals one if the respondent declared testing for HIV in the past year. Outcomes in Columns 6 and 7 are indices constructed following [Anderson \(2008\)](#). Column 6 is a combinaison of all the outcomes from Columns 1 to 4. Column 7 is constructed using information on the total number of sexual partners in their life, the number of sexual partners in the past month, and an indicator variable that equals one if they had unprotected sex in the past month. Panel A shows the average treatment effect and Panel B presents the heterogeneous effects by gender. All regressions include a strata fixed effect. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A9: Wikipedia Access and Long-Run Attitudes towards Sex, Gender Diversity, and Norms, Excluding Time Trend

	Definition Index	Acceptance Indices			
	(1)	(2) Same-sex relationships	(3) Transgender	(4) Gender Norms	(5) Sex Before Marriage
Panel A: Average Treatment Effect					
Treatment	-0.037 (0.066)	-0.007 (0.056)	0.109 (0.070)	0.028 (0.050)	-0.358 (0.340)
Panel B: Heterogeneous Treatment Effect					
Treatment $\times$ male	-0.002 (0.095)	-0.074 (0.079)	-0.017 (0.094)	-0.044 (0.069)	-0.702 (0.480)
Treatment $\times$ female	-0.081 (0.096)	0.081 (0.080)	0.270*** (0.103)	0.117 (0.077)	0.074 (0.483)
Male	-0.002 (0.118)	0.303*** (0.099)	0.371*** (0.121)	0.038 (0.110)	0.564 (0.679)
Mean control	-0.001	0.000	-0.001	0.000	4.625
Mean control - male	-0.021	0.073	0.082	0.030	4.935
Mean control - female	0.024	-0.092	-0.105	-0.038	4.236
Observations	510	510	510	510	507

Notes: This table shows the average and heterogeneous treatment effect of access to Wikipedia on long-run attitudes towards sexuality and gender, disaggregated by gender, without the time trend. Outcomes in Columns 1 to 4 are indices constructed following [Anderson \(2008\)](#). The outcome in Column 1 uses categorical variables on whether the respondent can correctly define LGBT and transgender. See Appendix Table A5 for examples of definitions provided by the respondents. The outcome in Column 2 include three distinct variables taking a value between 1 (never justified) and 10 (always justified) regarding homosexuality, homosexuality between two adult men, and two adult women, and two variables ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that homosexual activity should be not criminalized and whether same-sex marriage should be legalized. The outcome in Column 3 includes two distinct variables taking a value between 1 (never justified) and 10 (always justified) regarding a person born male who lives and dresses like a woman and a person born female who lives and dresses like a man, and one variable ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that a person born male who lives like a woman should be free to dress as a woman at work without facing discrimination. The outcome in Column 4 includes one variable taking a value between 1 (never justified) and 10 (always justified) regarding abortion, and two variables ranging between 1 (strongly disagree) and 5 (strongly agree) on whether they agree that men make better executives than women (reversed score) and whether a husband should help his wife with household chores. See Appendix Figure A6 for the distribution of these measures in the full sample. Finally, the outcome in Column 5 takes a value between 1 (never justified) and 10 (always justified) regarding sex before marriage. All regressions include a strata fixed effect. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A10: Wikipedia Access and Long-Run Internet Use, Excluding Time Trend

Panel A: Access to a Smartphone			
	(1) Had Phone Before 20	(2) Minutes on Phone	(3) *Has Digital Wellbeing*
Treatment	0.042 (0.040)	4.293 (21.585)	-0.004 (0.044)
Treatment $\times$ male	0.024 (0.057)	-13.771 (34.162)	-0.086 (0.058)
Treatment $\times$ female	0.060 (0.055)	20.413 (26.524)	0.097 (0.069)
Male	-0.238*** (0.076)	-50.301 (42.499)	0.015 (0.092)
Mean control	0.604	271.761	0.440
Mean control - male	0.518	257.630	0.424
Mean control - female	0.712	289.246	0.459
Observations	511	270	511
Panel B: Internet Use			
	Days on Internet	Internet Bundles	*Free Internet*
Treatment	-0.063 (0.206)	-1161.289* (692.996)	0.060 (0.042)
Treatment $\times$ male	-0.319 (0.301)	-794.657 (1014.002)	0.008 (0.057)
Treatment $\times$ female	0.245 (0.264)	-1653.848* (939.253)	0.125* (0.065)
Male	-0.384 (0.444)	1606.329 (1259.527)	0.022 (0.078)
Mean control	5.560	7020.014	0.402
Mean control - male	5.439	7130.219	0.403
Mean control - female	5.712	6868.333	0.400
Observations	510	442	509
Panel C: Wikipedia Use			
	Used Wikipedia	Wikipedia Pages	*Has Browsing History*
Treatment	0.083*** (0.027)	0.089 (0.814)	-0.016 (0.041)
Treatment $\times$ male	0.061* (0.034)	0.949 (0.673)	-0.031 (0.058)
Treatment $\times$ female	0.113*** (0.042)	-0.541 (1.319)	-0.003 (0.054)
Male	0.067 (0.059)	-3.907*** (1.079)	-0.323*** (0.080)
Mean control	0.855	5.130	0.644
Mean control - male	0.885	2.486	0.518
Mean control - female	0.818	7.270	0.802
Observations	510	325	511
Panel D: Social Media, Wikipedia, and Health			
	Social Media Usage	Social Media and Health	Wikipedia and Health
Treatment	0.003 (0.036)	-0.053 (0.052)	0.076 (0.066)
Treatment $\times$ male	-0.033 (0.047)	-0.032 (0.068)	0.115 (0.090)
Treatment $\times$ female	0.050 (0.057)	-0.081 (0.081)	0.018 (0.096)
Male	0.085 (0.071)	-0.103 (0.106)	-0.466*** (0.132)
Mean control	0.021	-0.000	-0.001
Mean control - male	-0.007	-0.035	-0.132
Mean control - female	0.057	0.043	0.165
Observations	510	510	510

Notes: This table shows the average and heterogeneous treatment effect of access to Wikipedia on long-run internet use, disaggregated by gender, without the time trend. Panel A. Access to smartphone: (1) Owned phone before 20 years old; (2) Minutes on smartphone last Wednesday/Saturday; (3) Has Digital Wellbeing Panel B. Internet usage: (1) Days they used the internet over the past week; (2) Amount spent in Malawian Kwacha on internet bundles; (3) Accessed internet for free in the past month. Panel C. Wikipedia usage: (1) Ever used Wikipedia; (2) Number of Wikipedia pages they looked at in the past week; (3) Have browsing history. Panel D. Social media usage and online information on health. All indices are constructed following [Anderson \(2008\)](#): (1) Amount spent on social media bundles, minutes using social media last Wednesday/Saturday, how often they used social media in the past month, and have social media accounts (Facebook, Twitter, Tiktok, and WhatsApp); (2) Use social media for information about health and social media in top 3 best place to find information about health; (3) Use Wikipedia for information about health and Wikipedia in top 3 best place to find information about health. Outcomes in Column 3 from Panels A, B, and C were not pre-specified in the pre-analysis plan. All regressions include a strata fixed effect and a time trend indicating the month-year of the long-run follow-up survey. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix Table A11: Is Wikipedia Information Better than the Internet?

	How is Wikipedia better		
	(1) Trustworthy	(2) Easy to understand	(3) Easy to find
Treatment	0.262*** (0.039)	0.333*** (0.038)	0.247*** (0.038)
Mean Control	0.436	0.495	0.542
Observations	549	549	549

Notes: From [Derksen et al. \(2022\)](#). The outcomes are indicators calculated based on answers to the question “How is Wikipedia better than other sites on the internet?”. (1) is equal to one if option “Information on Wikipedia is more trustworthy” was chosen. (2) if “It is easier to understand information on Wikipedia” was chosen. (3) if “It is easier to find information on Wikipedia” was chosen. Other alternatives were “There is more information on Wikipedia”, “There are more things to do on Wikipedia”, and “Don’t know”. The sample is students in the treatment group and in the subsample of the control group with supplementary surveys. All regressions include strata fixed effects. Randomization was stratified by school, form, above median achievement and past internet use. Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.